VOLTAGE-DIVIDER

BJT DC BIASING

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Thevenin's Theorem

- Thevenin Voltage
- Thevenin Resistance

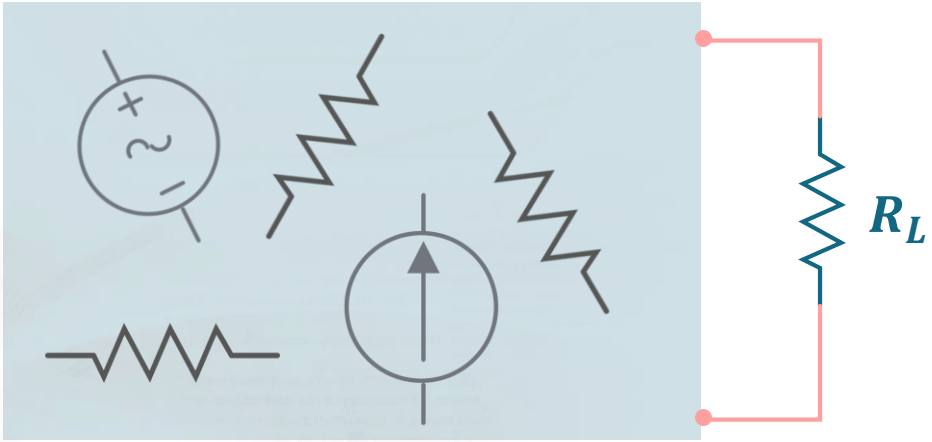
Voltage-Divider Bias Circuit

- Base-Emitter Loop
- Collector-Emitter Loop
- Load Line Analysis

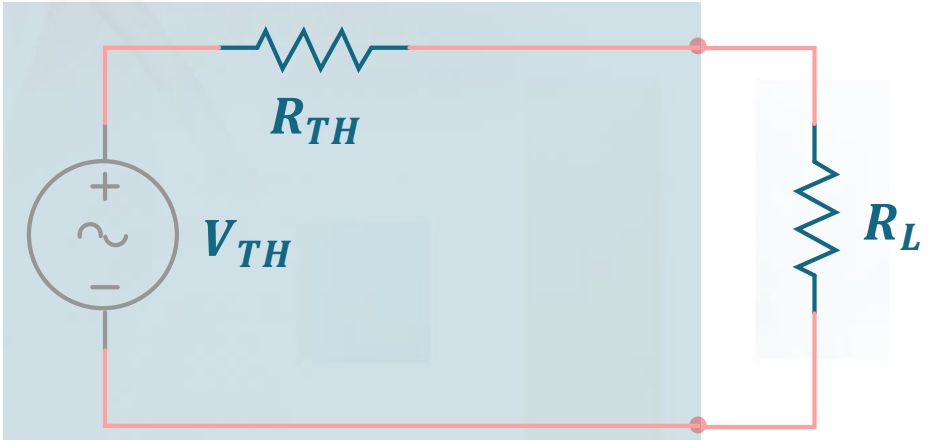
THEVENIN'S THEOREM

It is possible to simplify any **linear circuit**, irrespective of how complex it is, to an equivalent circuit with a single voltage source, V_{TH} and a series resistance, R_{TH}.

Arbitrary Network

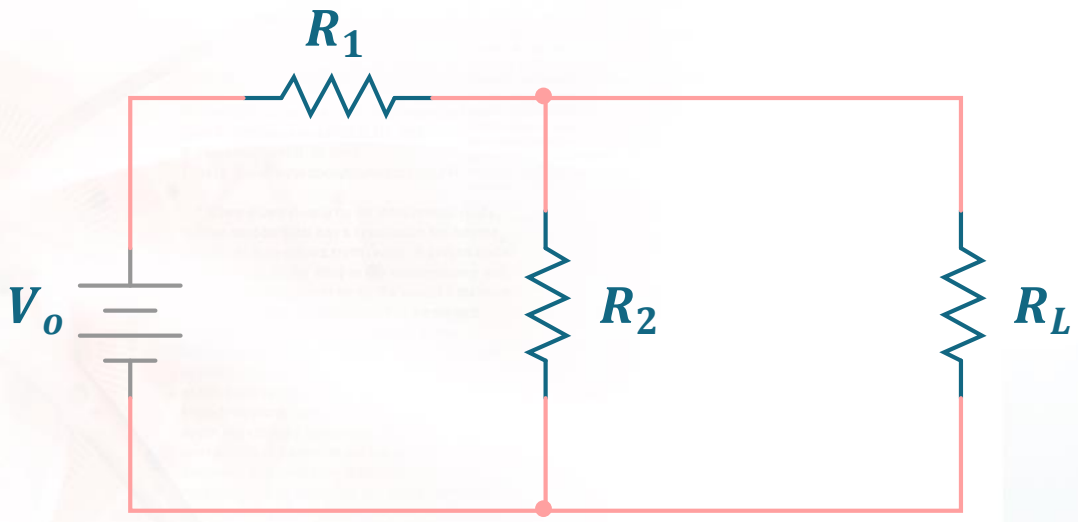


Thevenin Equivalent



THEVENIN'S VOLTAGE, V_{TH}

The **open-circuit voltage**—it represents the voltage across two terminals if **no load** were connected



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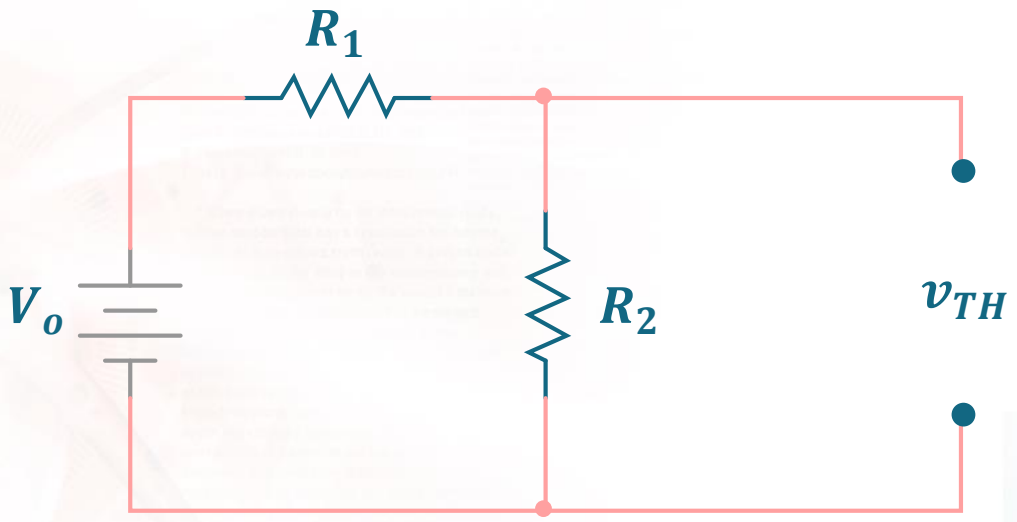
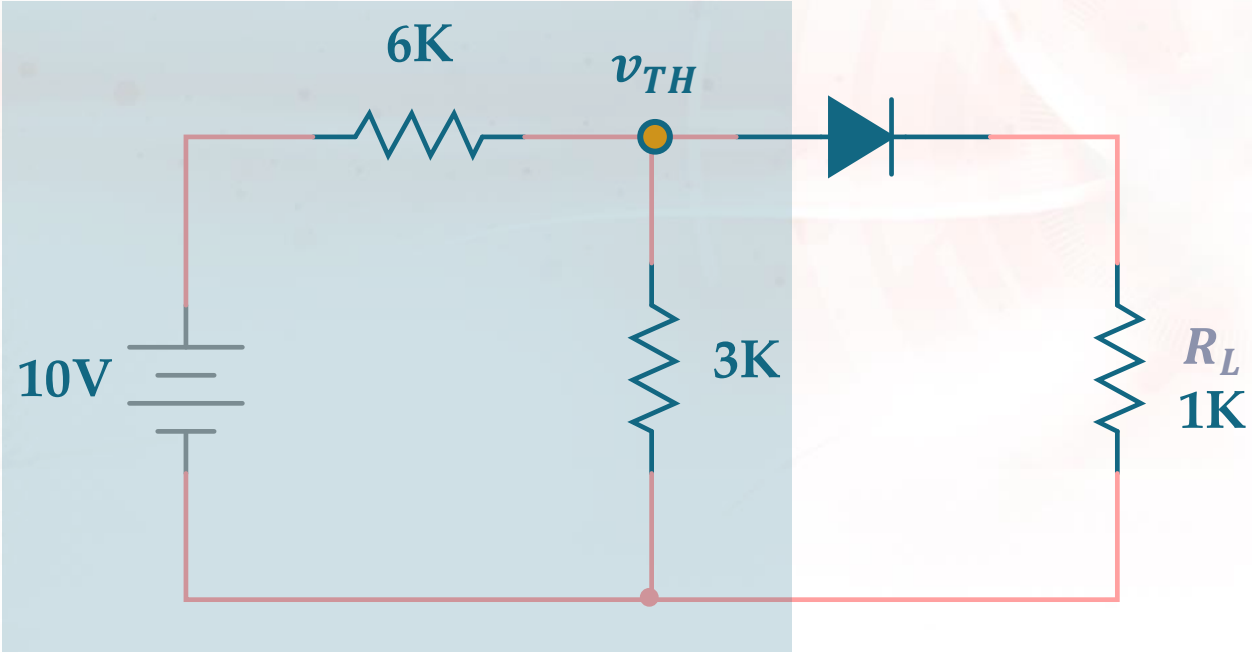


Determine the **Thevenin voltage** of the given network.

Thevenin Equivalent Circuit



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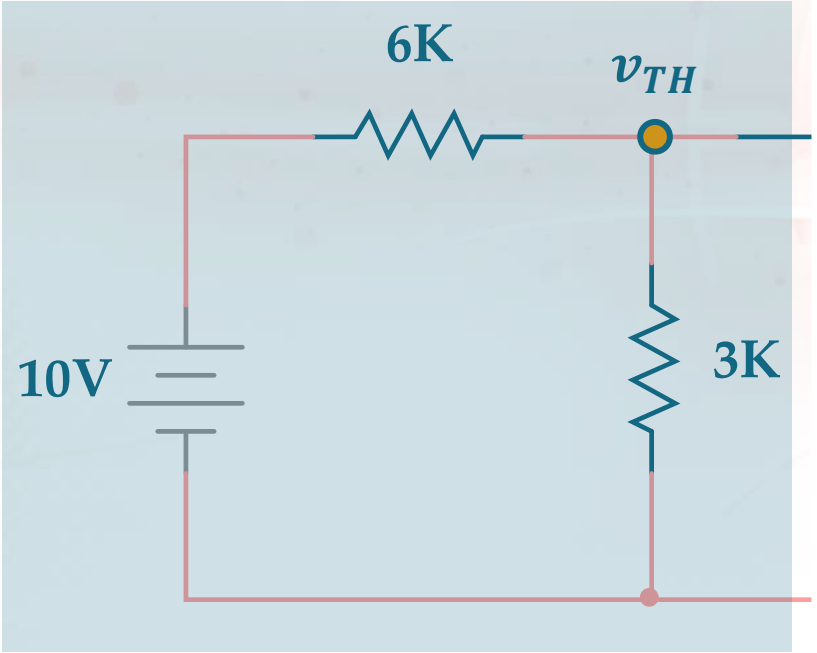


Voltage Divider Theorem, VDT

$$v_{R_2} = v_o \frac{R_2}{R_o}$$

$$v_{TH} = 10 \frac{3K}{3K + 6K}$$

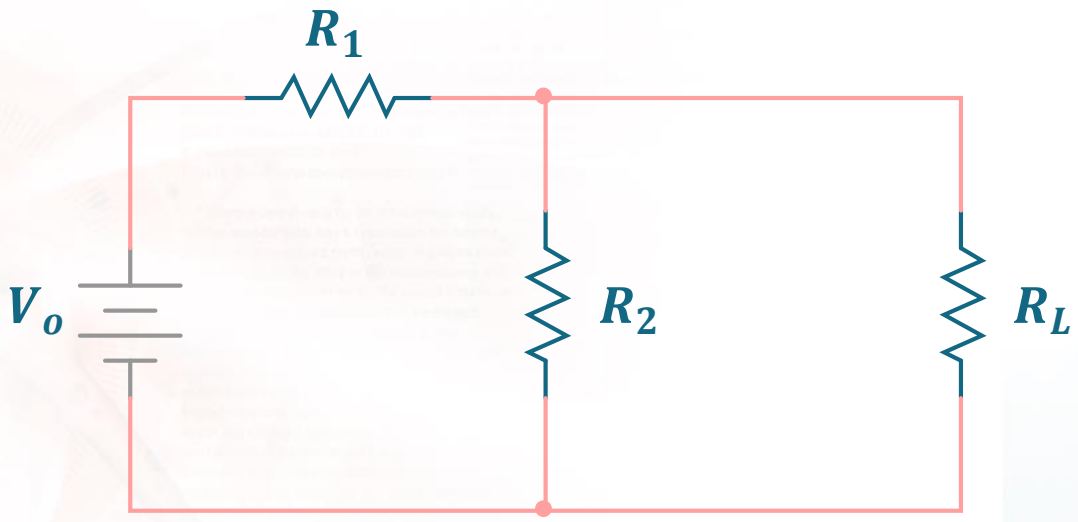
$$\boxed{v_{TH} = 3.33\ V}$$

ans

THEVENIN'S RESISTANCE, R_{TH}

The equivalent resistance across the **open load** terminals when all independent **sources** are set to **zero**

- voltage source → short-circuit
- current source → open-circuit



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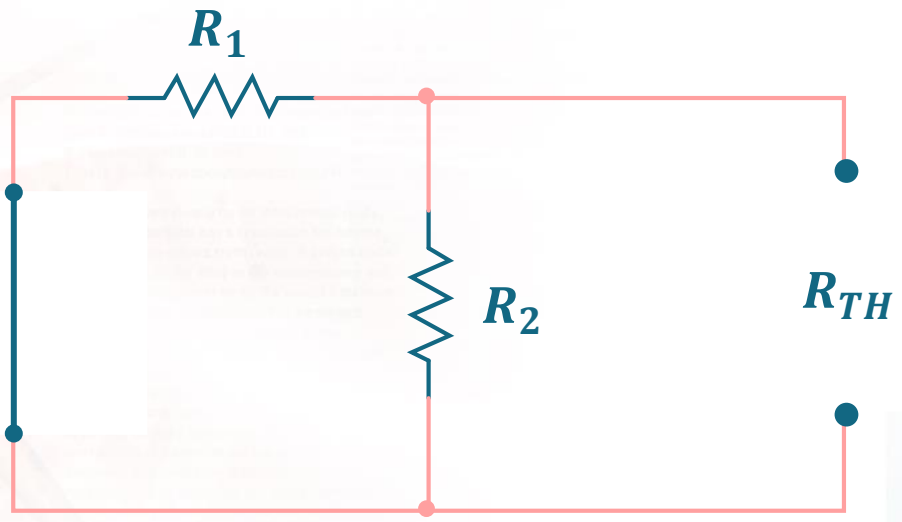

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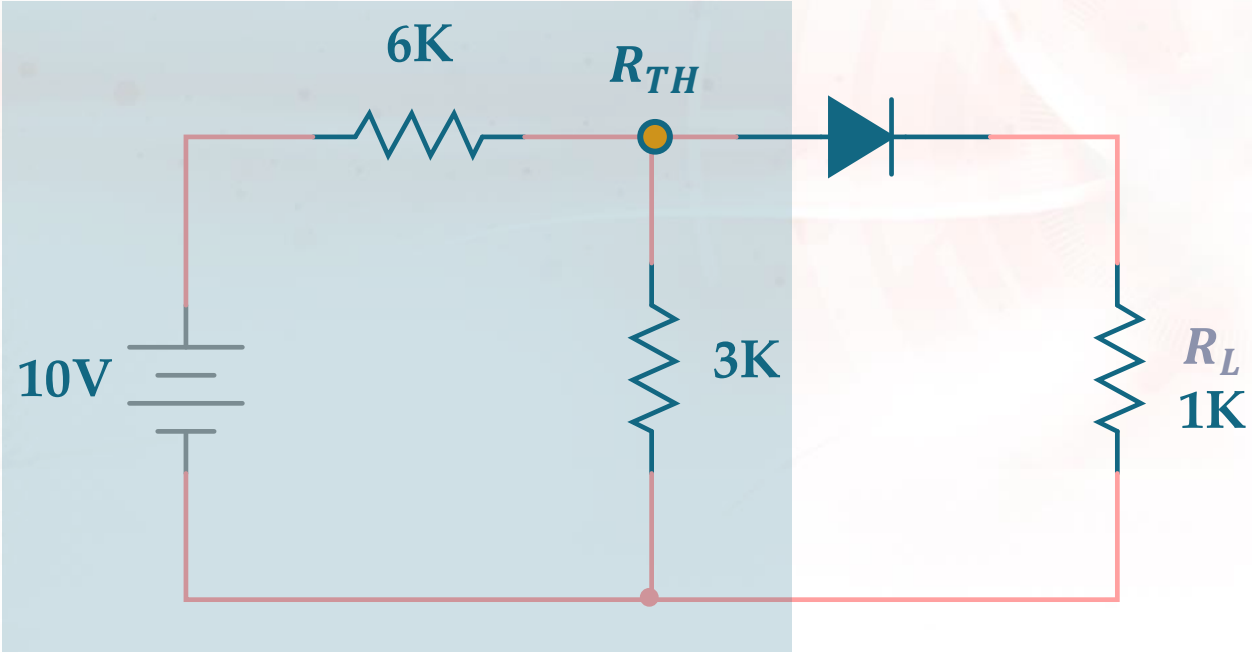

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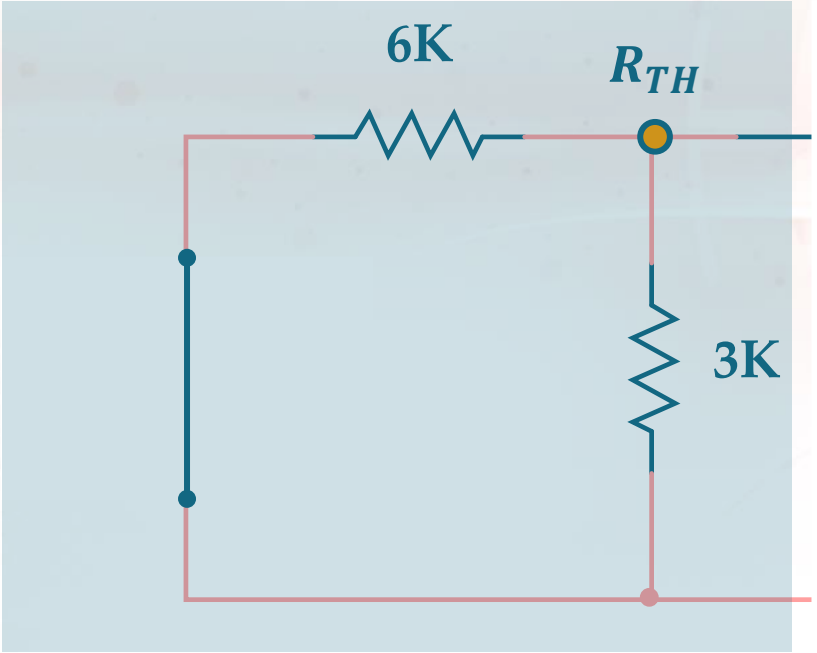


3K is in parallel with 6K

$$\frac{1}{R_{TH}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{TH}} = \frac{1}{6K} + \frac{1}{3K}$$

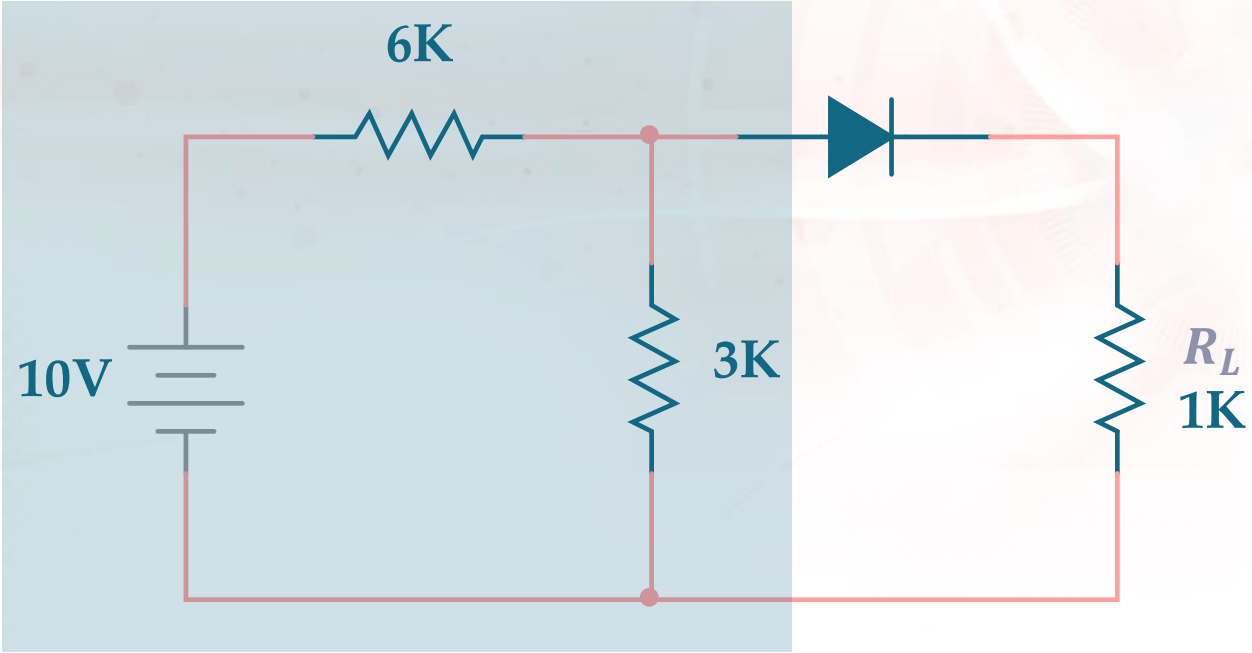
$$\frac{1}{R_{TH}} = \frac{1}{2K}$$

$$\boxed{R_{TH} = 2K}$$

ans

Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.

Thevenin Equivalent Circuit



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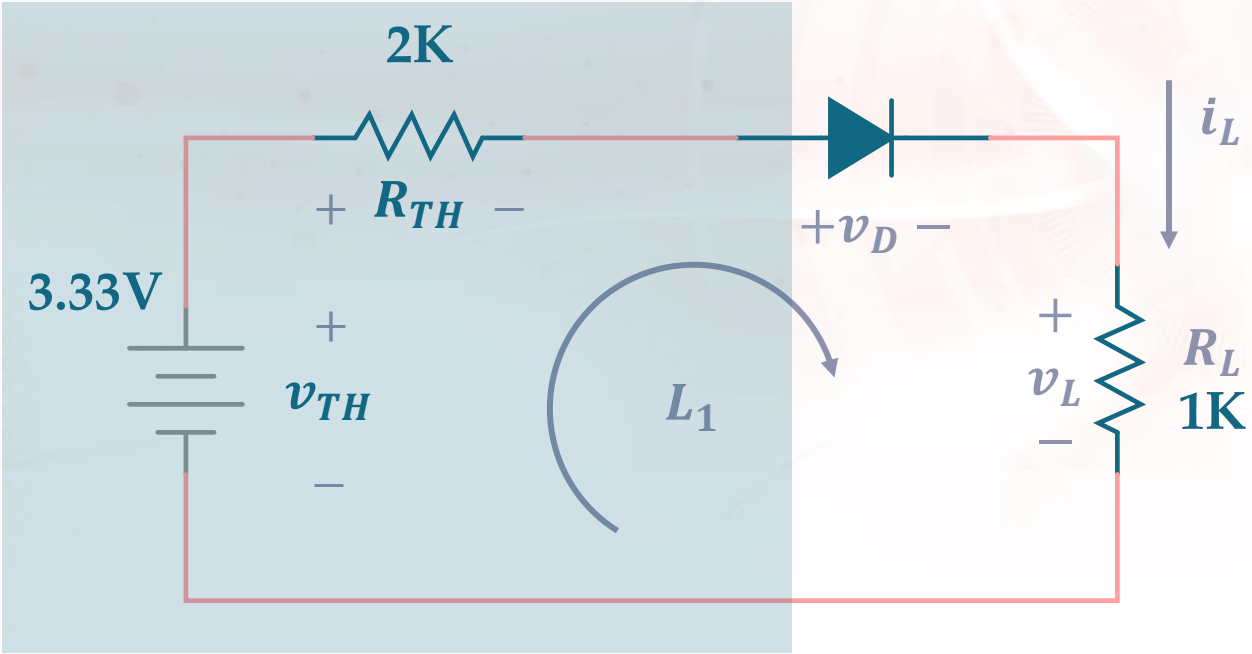

KVL @L_1

$$-v_{TH} + v_{RTH} + v_D + v_L = 0$$

$$v_{RTH} + v_L = 3.33 - 0.7$$

$$i_L R_{TH} + i_L R_L = 2.63$$

$$i_L(R_{TH} + R_L) = 2.63$$

$$i_L = \frac{2.63}{2K + 1K}$$

$$\boxed{i_L = 876.67\ uA}$$

ans

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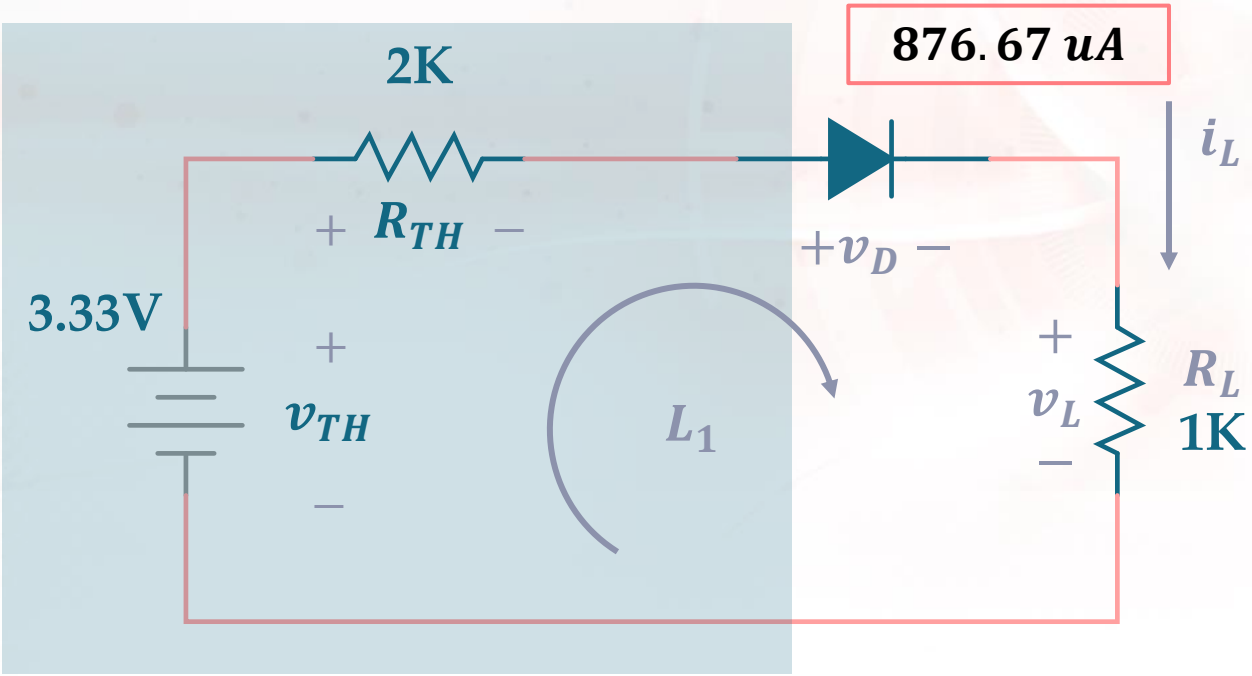
Thevenin Equivalent Circuit



Ohm's Law

$$v_L = i_L R_L$$

$$v_L = 876.67\ uA(1K)$$

$$\boxed{v_L = 0.88\ V}$$

ans

VOLTAGE-DIVIDER BIAS CIRCUIT

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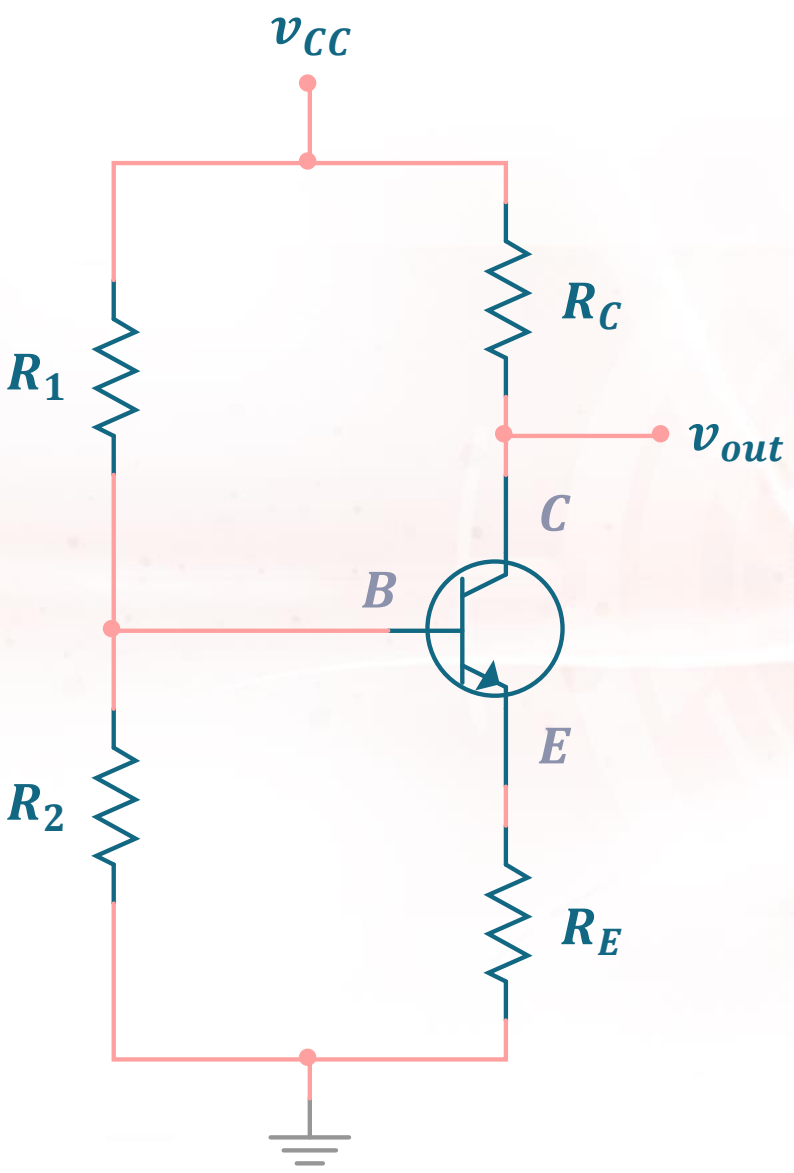
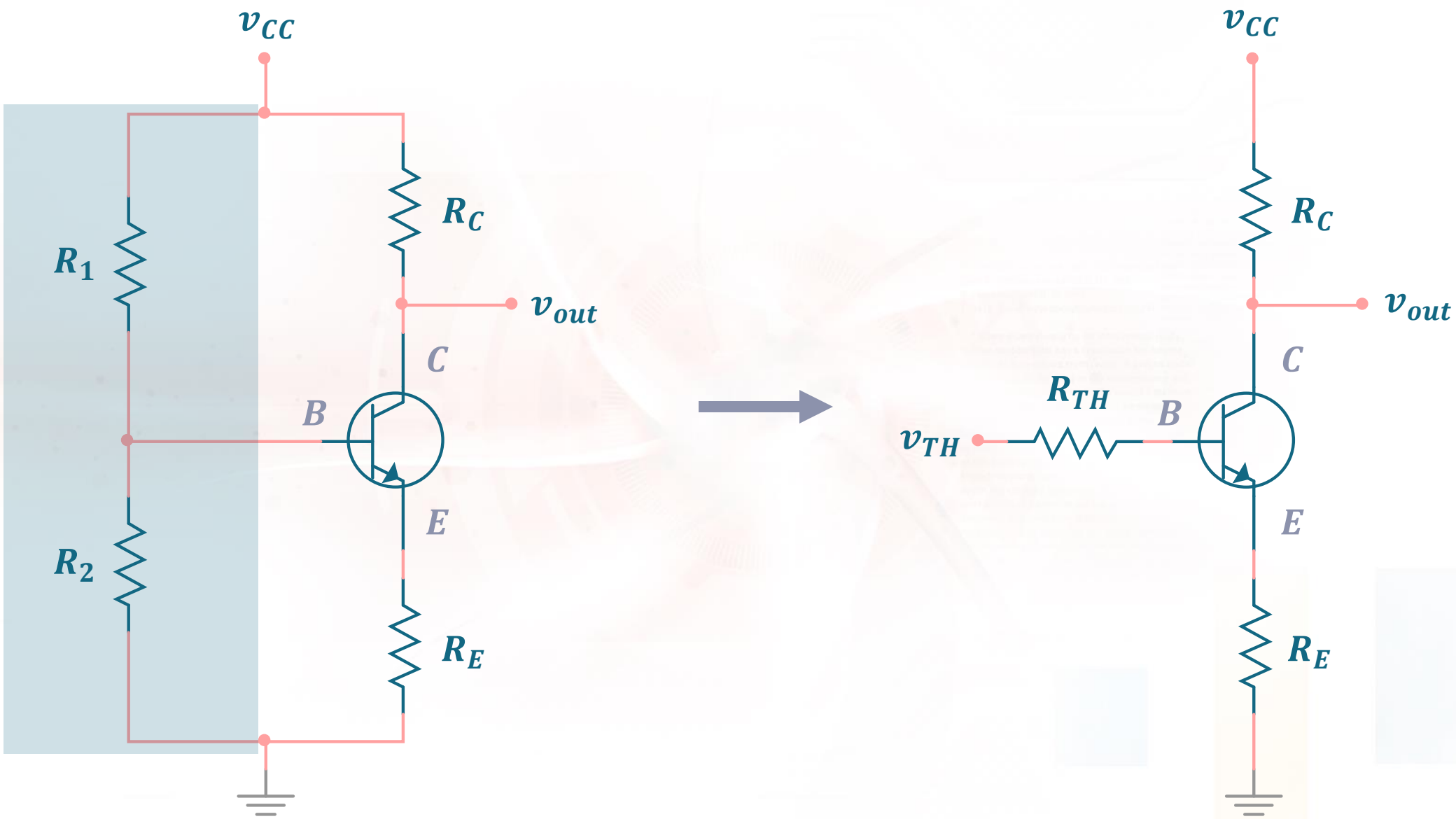
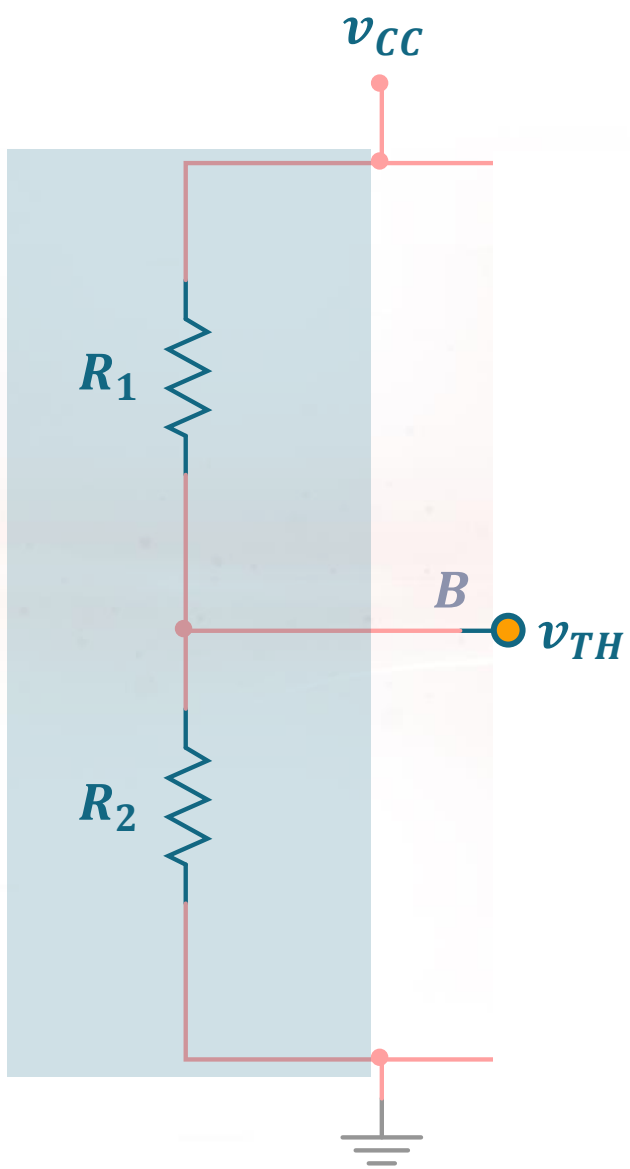


THEVENIN EQUIVALENT CIRCUIT



THEVENIN VOLTAGE, v_{TH}



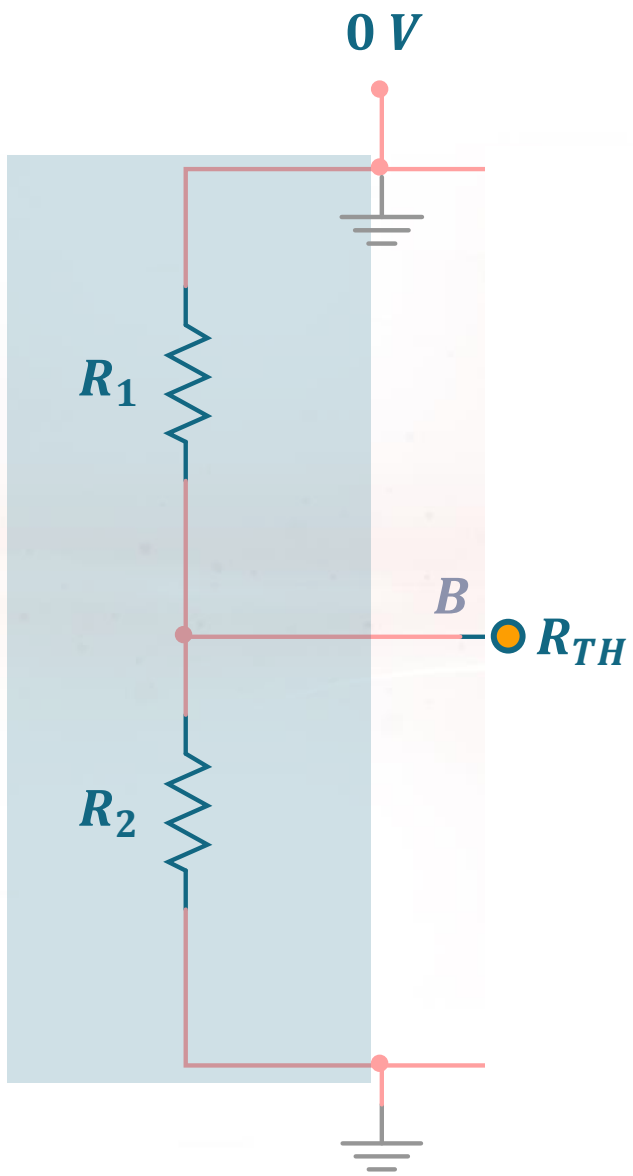
Voltage Divider Theorem, VDT

$$v_{R2} = v_o \frac{R_2}{R_o}$$

$$v_{TH} = v_{CC} \frac{R_2}{R_1 + R_2}$$

Voltage-Divider Bias

THEVENIN RESISTANCE, R_{TH}



R_1 is in parallel with R_2

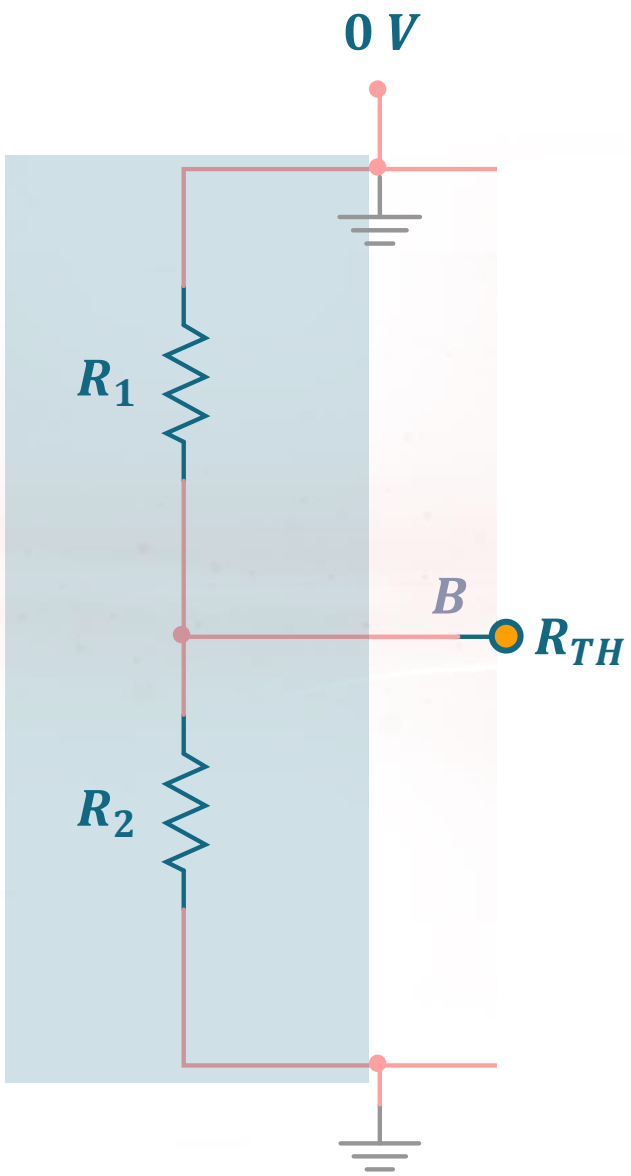
$$\frac{1}{R_{TH}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{TH}} = \frac{1}{R_1} \cdot \frac{R_2}{R_2} + \frac{1}{R_2} \cdot \frac{R_1}{R_1}$$

$$\frac{1}{R_{TH}} = \frac{R_2 + R_1}{R_1 R_2}$$

$$R_{TH} = \frac{R_1 R_2}{R_1 + R_2}$$

THEVENIN RESISTANCE, R_{TH}



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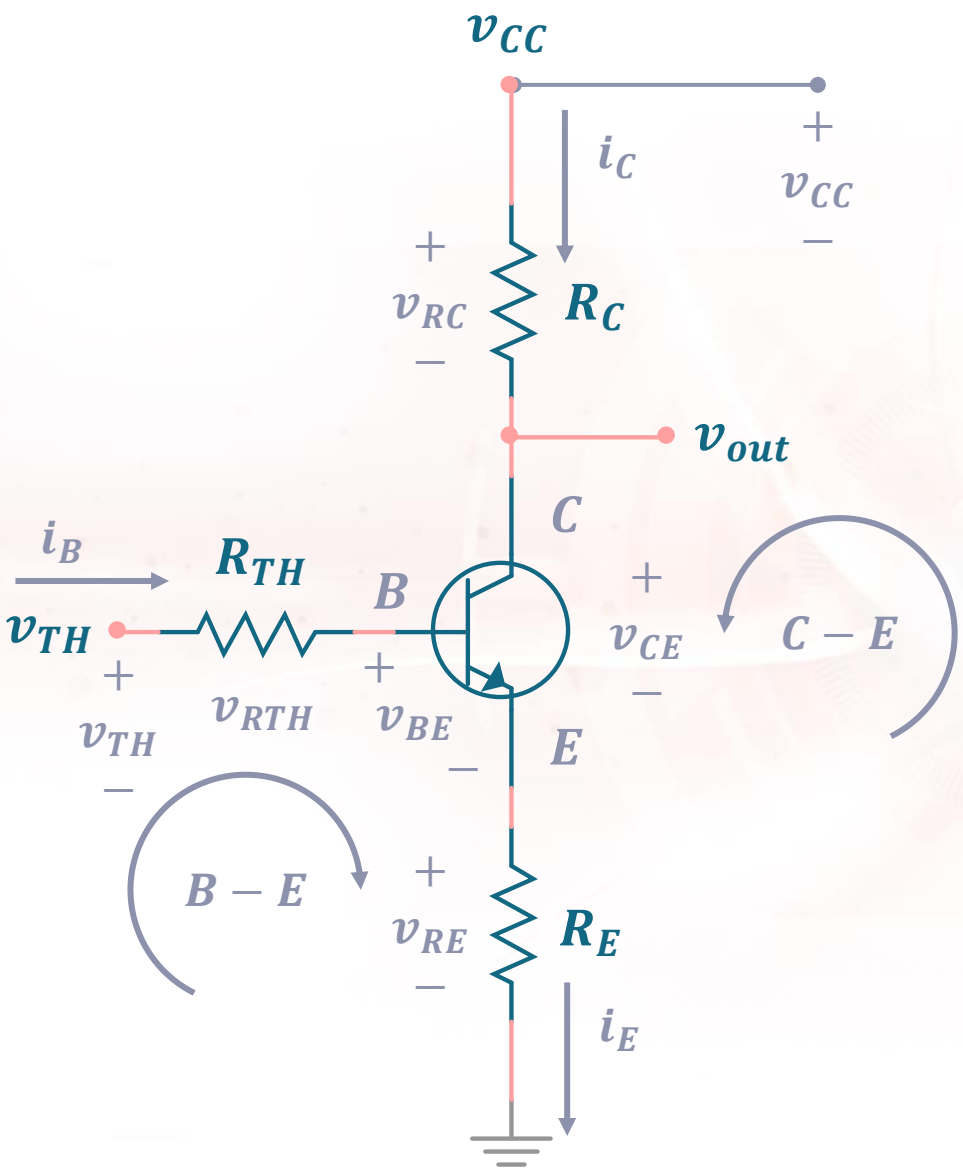
$$\frac{1}{R_{TH}} = \frac{R_2 + R_1}{R_1 R_2}$$

$$R_{TH} = \frac{R_1 R_2}{R_1 + R_2}$$

Voltage-Divider Bias

BASE-EMITTER LOOP



KVL @$B-E$

$$-v_{TH}+v_{RTH}+v_{BE}+v_{RE}=0$$

$$v_{RTH}+v_{RE}=v_{TH}-v_{BE}$$

$$i_BR_{TH}+i_ER_E=v_{TH}-v_{BE}$$

$$i_BR_{TH}+(\beta+1)i_BR_E=v_{TH}-v_{BE}$$

$$i_B[R_{TH}+(\beta+1)R_E]=v_{TH}-v_{BE}$$

$$i_B=\frac{v_{TH}-v_{BE}}{R_{TH}+(\beta+1)R_E}$$

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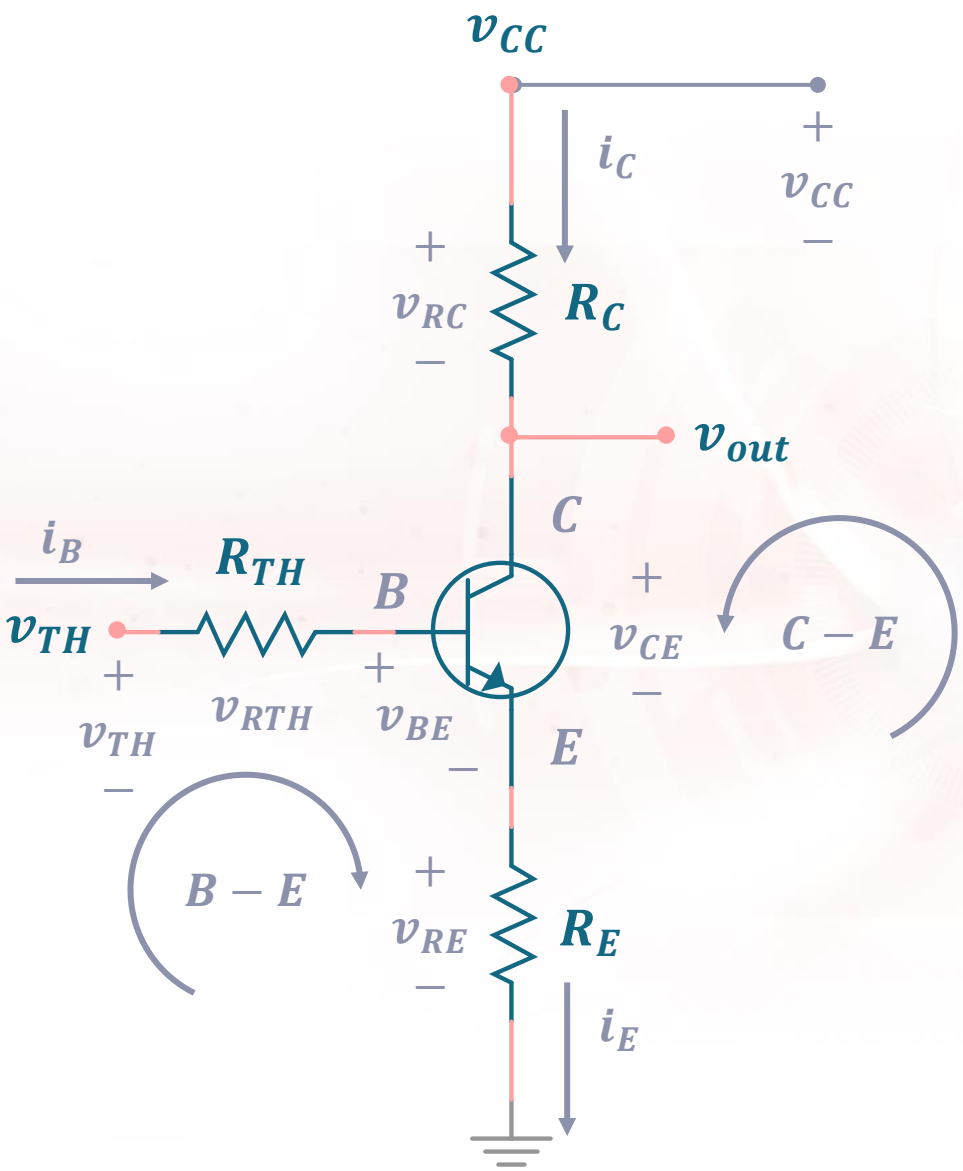

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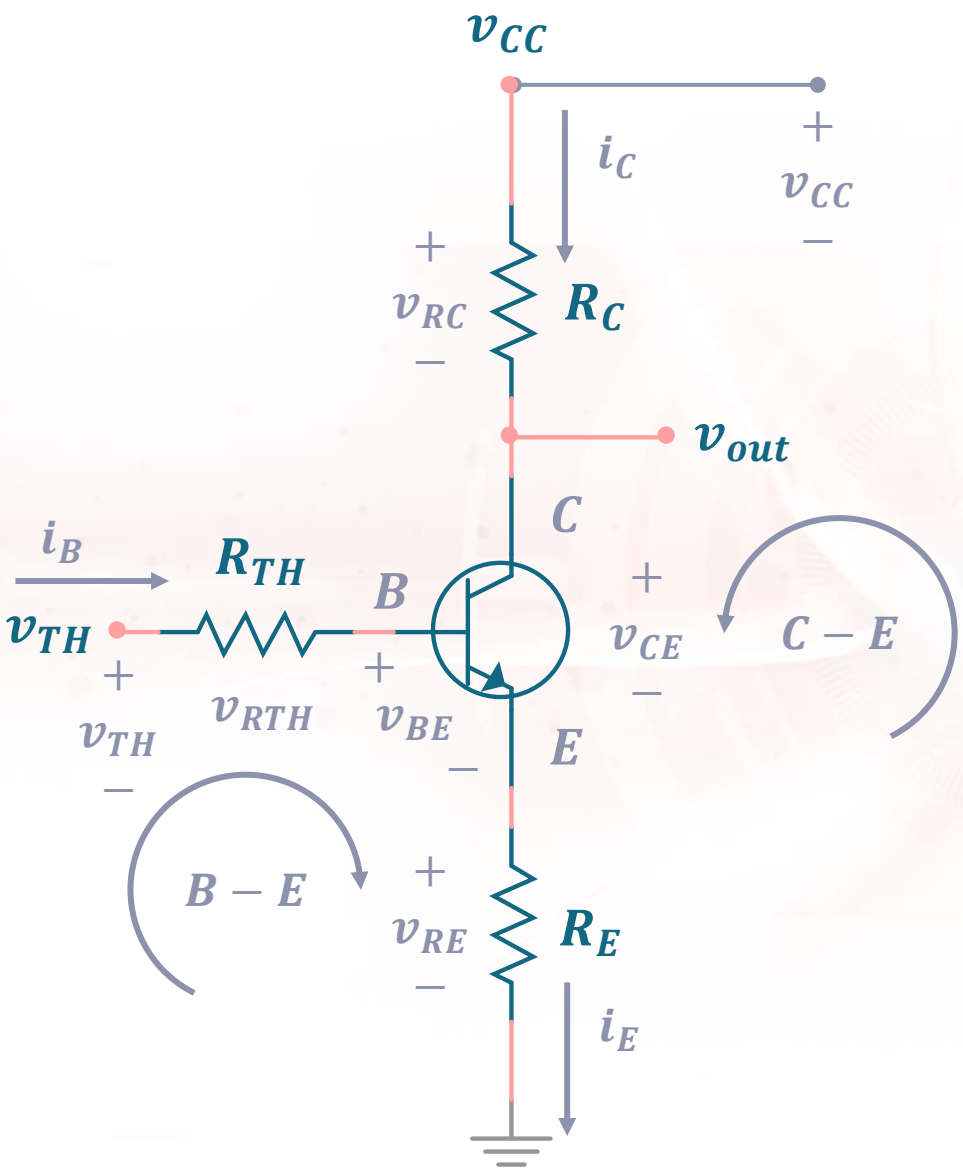
Voltage-Divider Bias

COLLECTOR-EMITTER LOOP



KVL @$C-E$

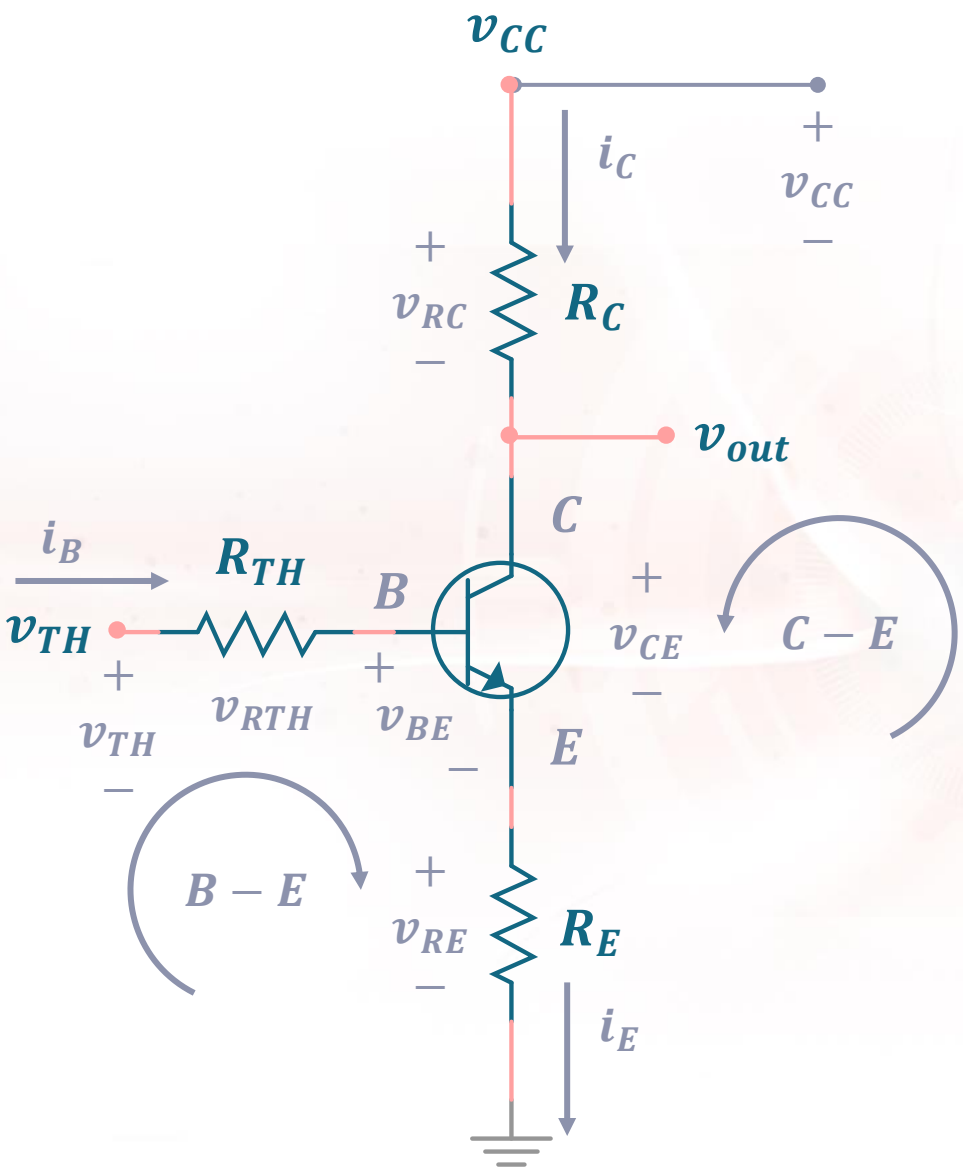
$$-v_{CC} + v_{RC} + v_{CE} + v_{RE} = 0$$

$$v_{CE} = v_{CC} - v_{RC} - v_{RE}$$

$$v_{CE} = v_{CC} - i_C R_C - i_E R_E$$

$$v_{CE} = v_{CC} - i_C(R_C + R_E)$$

COLLECTOR-EMITTER LOOP



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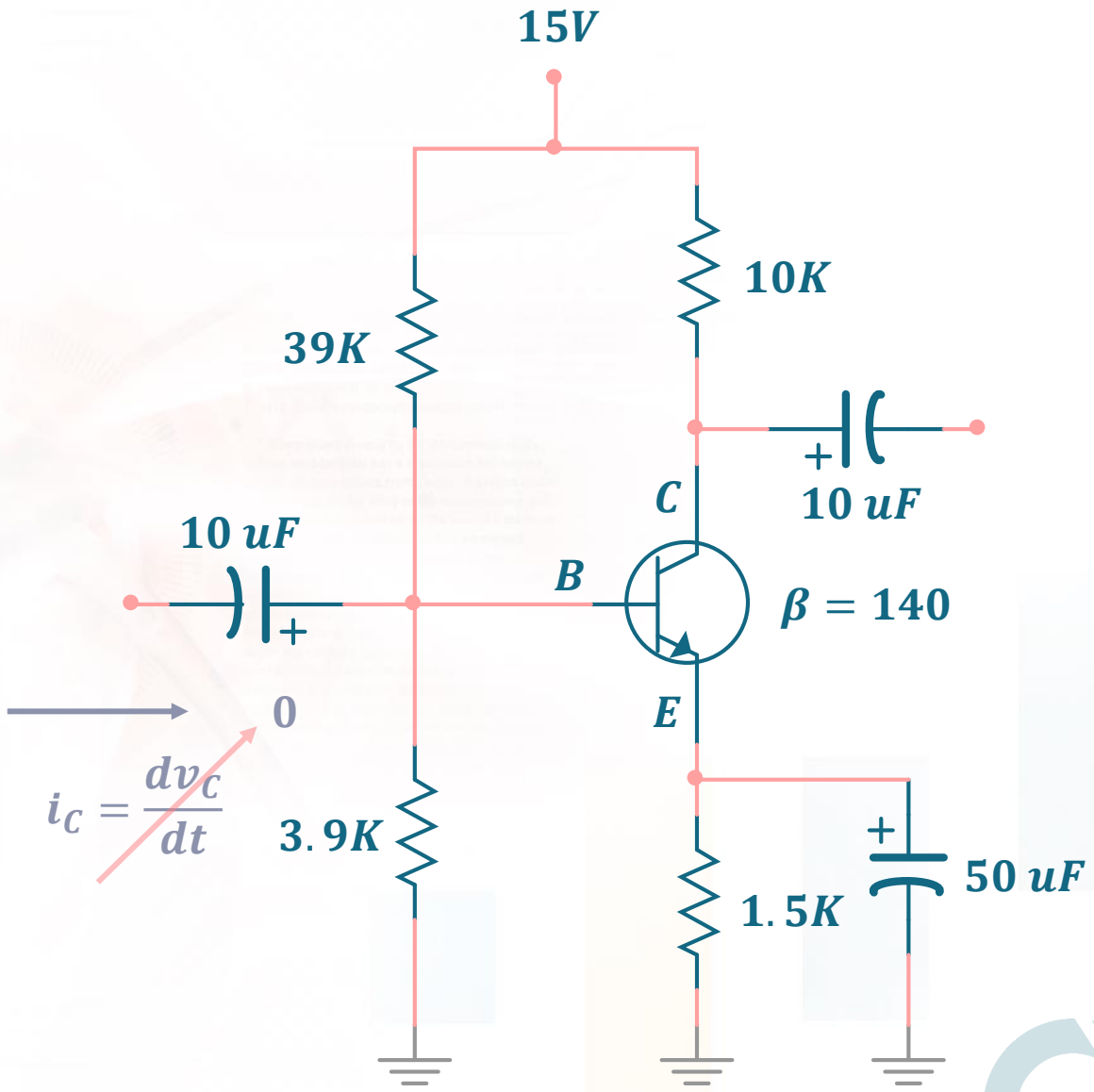
$$v_{CE} = v_{CC} - i_C R_C - i_E R_E$$

$$v_{CE} = v_{CC} - i_C(R_C + R_E)$$

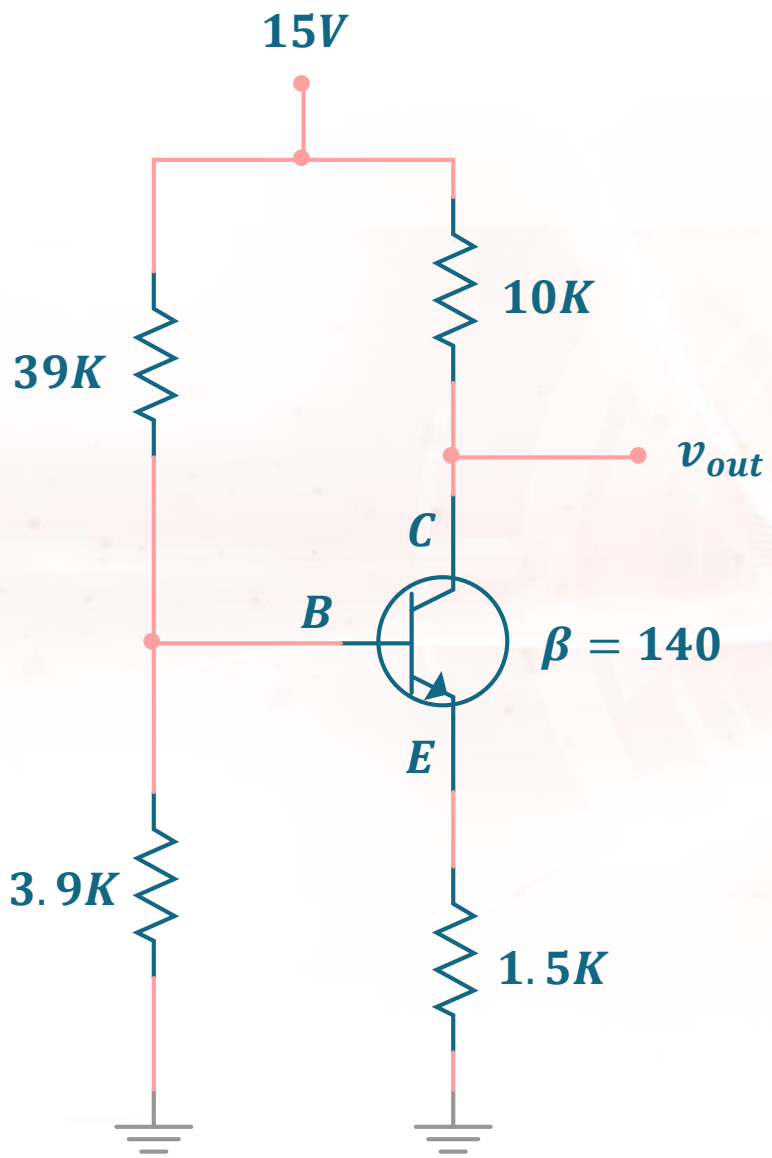
Voltage-Divider Bias

For the given **voltage-divider bias** network, determine:

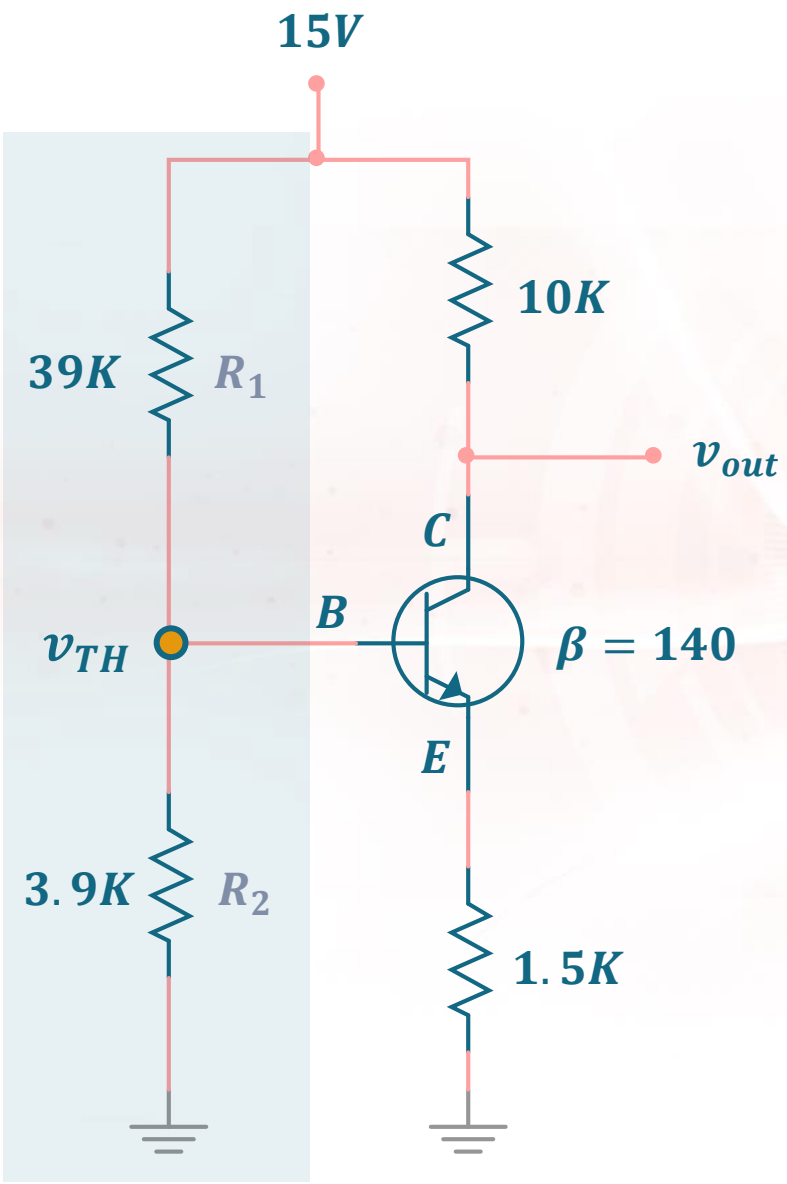
- Base current, i_B
- Collector current, i_{CQ}
- Collector-Emitter voltage, v_{CE}
- Emitter voltage, v_E
- Load Line Analysis
  - Saturation current, $i_{C(sat)}$
  - Cutoff voltage, $v_{CE(cutoff)}$



EXERCISE



THEVENIN VOLTAGE, v_{TH}-



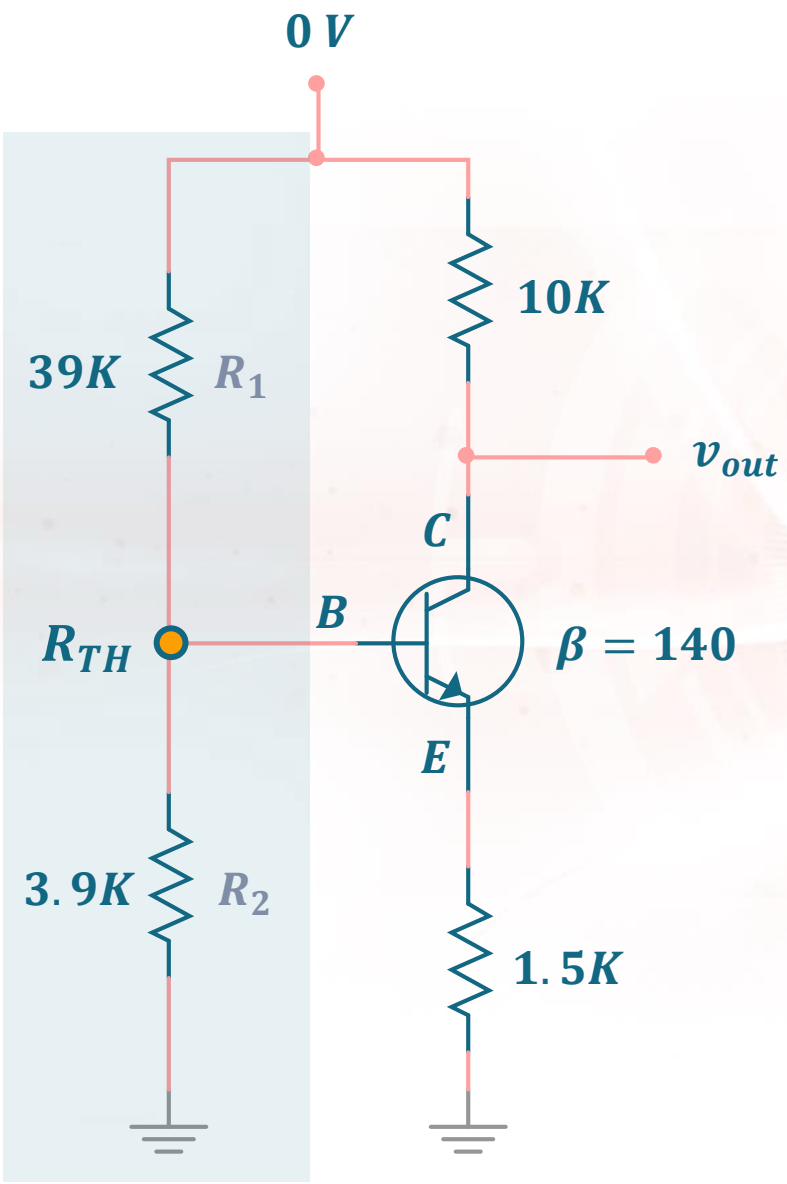
Voltage Divider Theorem, VDT

$$v_{TH} = v_{CC}\frac{R_2}{R_1 + R_2}$$

$$v_{TH} = 15\frac{39K}{39K + 3.9K}$$

$$v_{TH} = 1.36\ V$$

THEVENIN RESISTANCE, R_{TH}-



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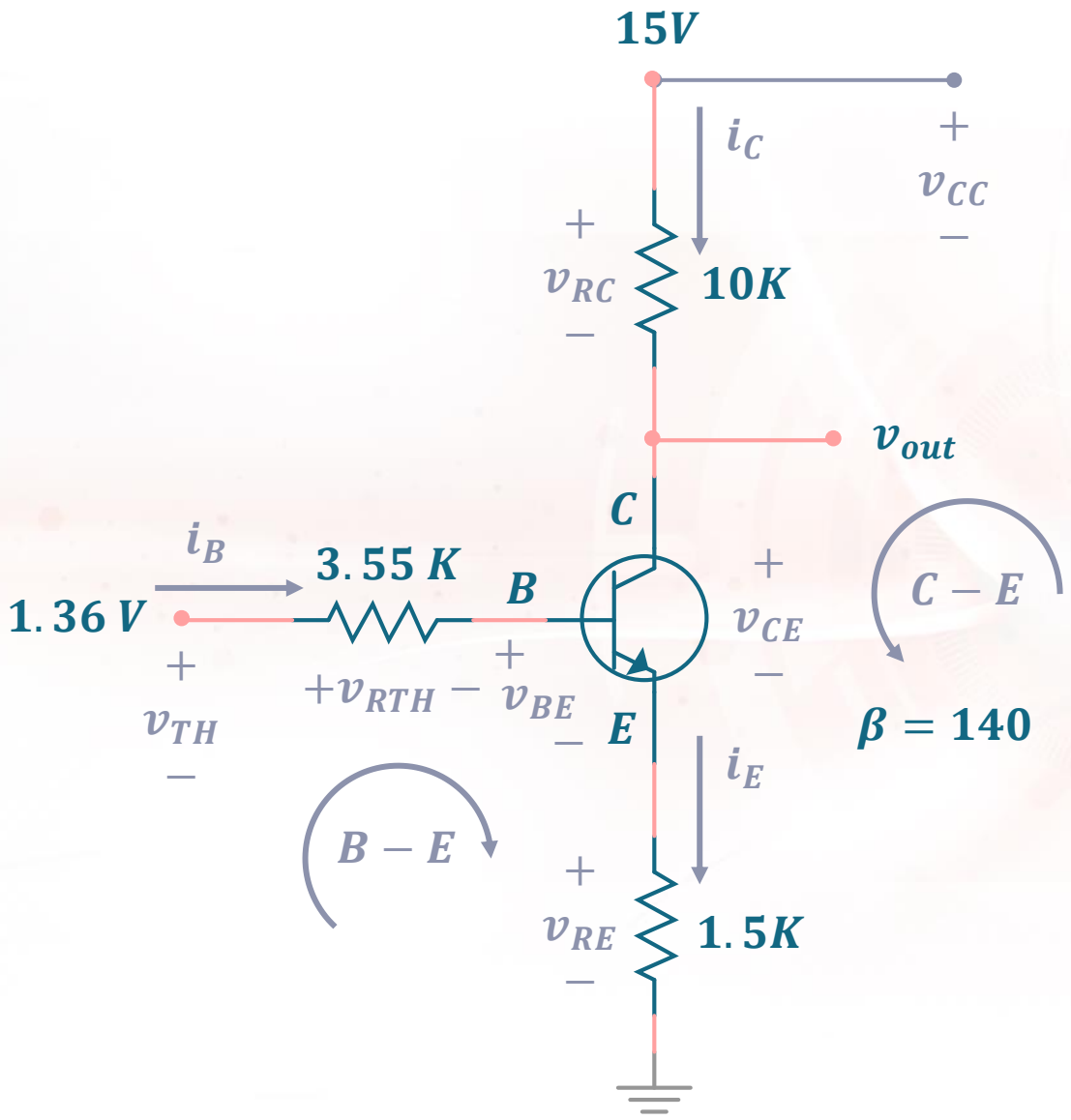
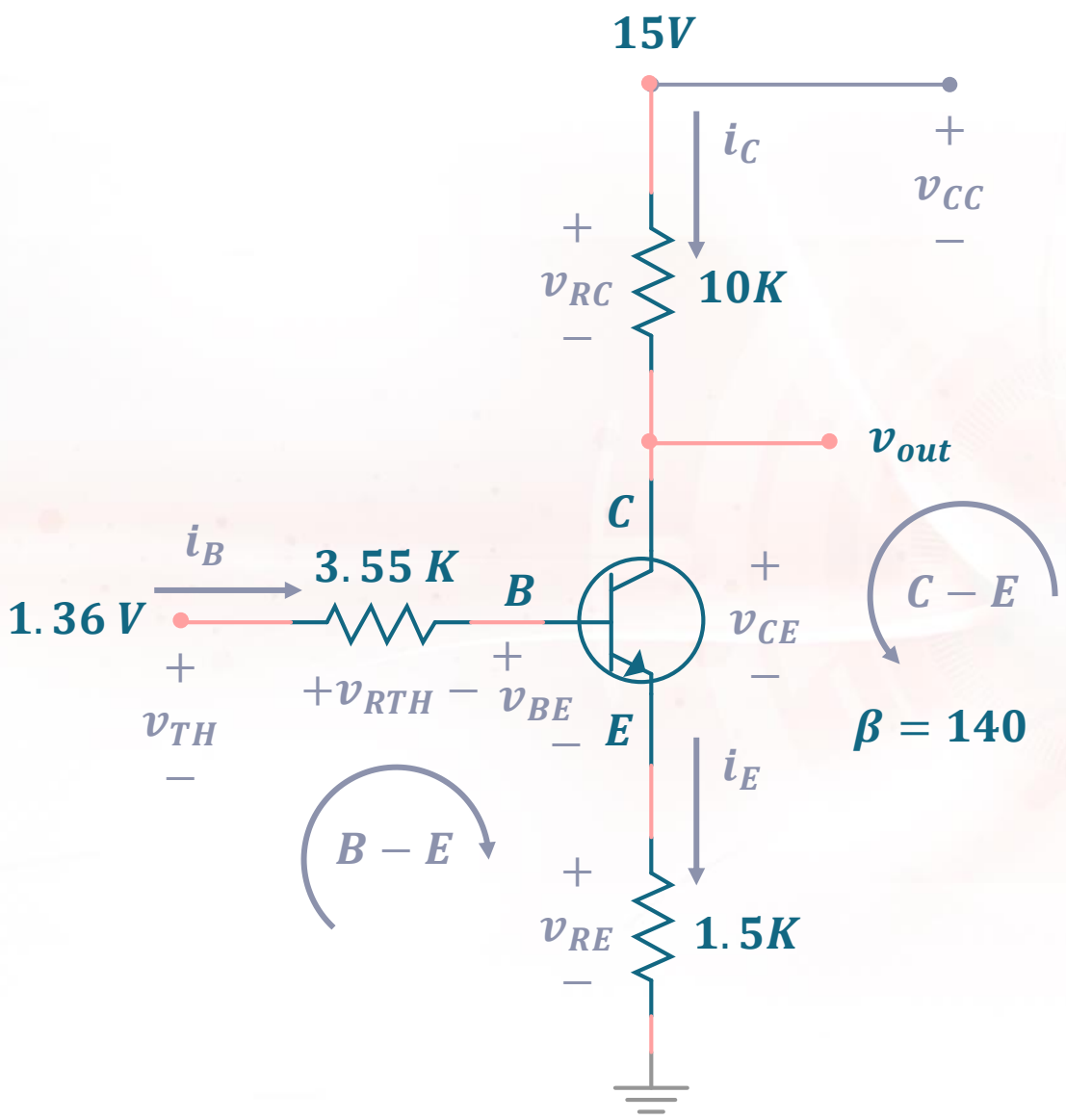
$$\frac{1}{R_{TH}} = \frac{1}{39K} + \frac{1}{3.9K}$$

$$\frac{1}{R_{TH}} = \frac{11}{39K}$$

$$R_{TH} = 3.55K$$

EXERCISE


15V
i_C
+
v_{CC}
−
+
v_{RC}
−
10K
v_{out}
C
i_B
3.55 K
B
1.36 V
+
v_{CE}
−
C − E
+
v_{TH}
−
+v_{RTH} −
+
v_{BE}
−
E
β = 140
i_E
B − E
+
v_{RE}
−
1.5K



KVL @$B - E$

$$-v_{TH} + v_{RTH} + v_{BE} + v_{RE} = 0$$

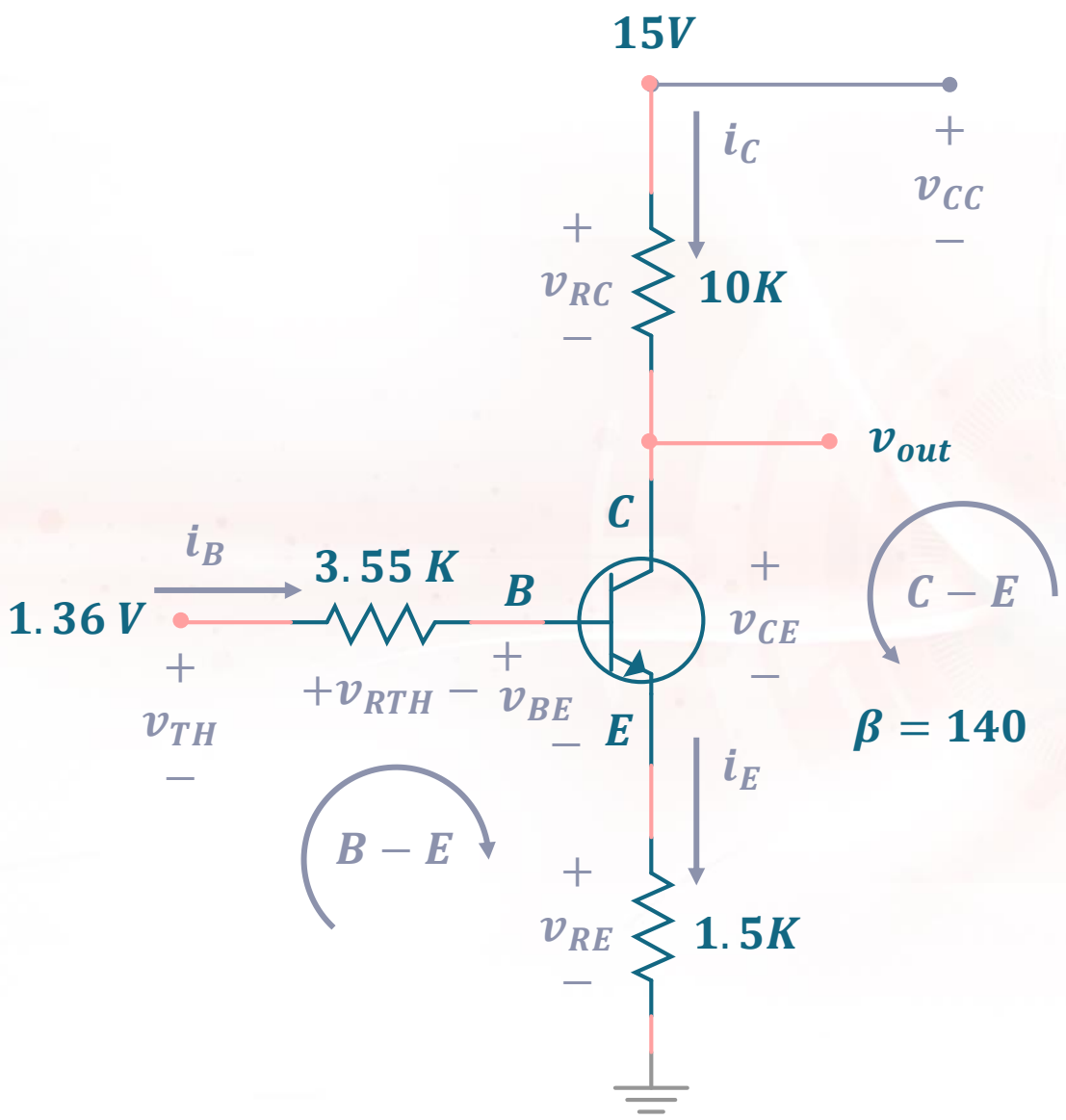
$$v_{RTH} + v_{RE} = v_{TH} - v_{BE}$$

$$i_B R_{TH} + i_E R_E = v_{TH} - v_{BE}$$

$$i_B R_{TH} + (\beta + 1) i_B R_E = v_{TH} - v_{BE}$$

$$i_B [R_{TH} + (\beta + 1) R_E] = v_{TH} - v_{BE}$$

$$i_B = \frac{v_{TH} - v_{BE}}{R_{TH} + (\beta + 1) R_E}$$



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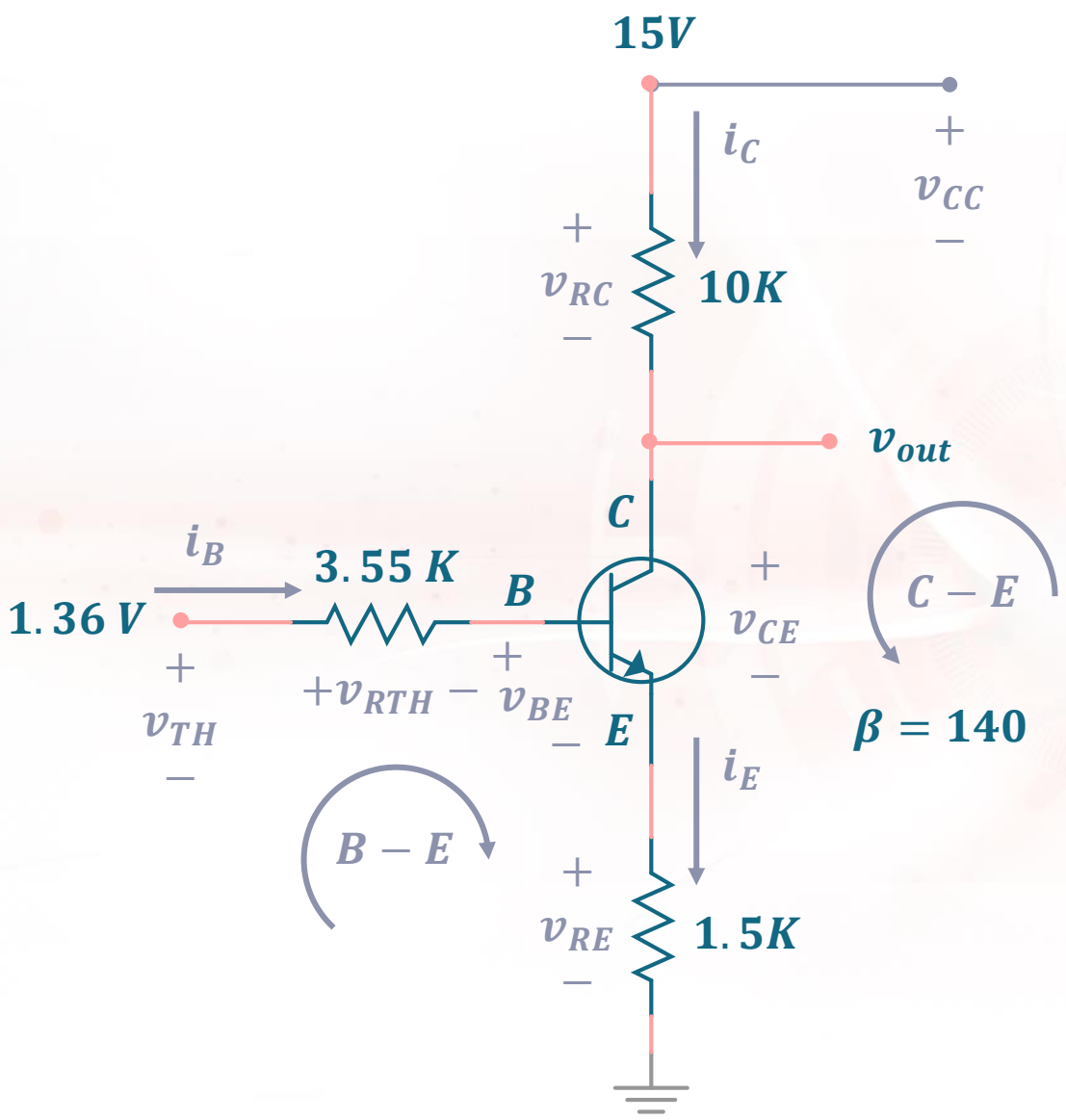
$$i_B = \frac{v_{TH} - v_{BE}}{R_{TH} + (\beta + 1)R_E}$$



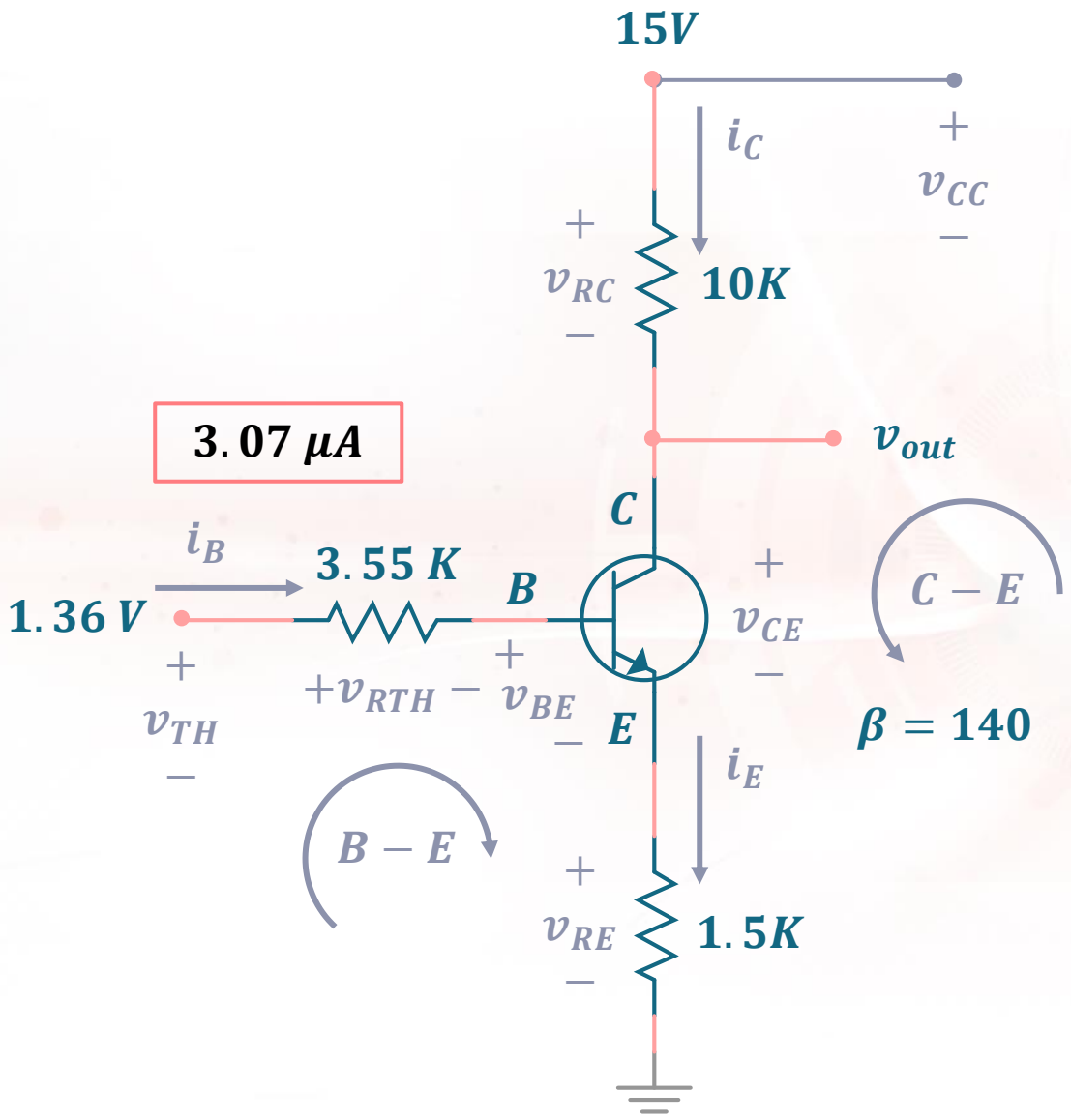
KVL @$B - E$

$$i_B = \frac{v_{TH} - v_{BE}}{R_{TH} + (\beta + 1)R_E}$$

$$i_B = \frac{1.36 - 0.7}{3.55K + (140 + 1)1.5K}$$

$$\boxed{i_B = 3.07\ \mu A}$$

ans



$\boxed{3.07\ \mu A}$

Current Gain

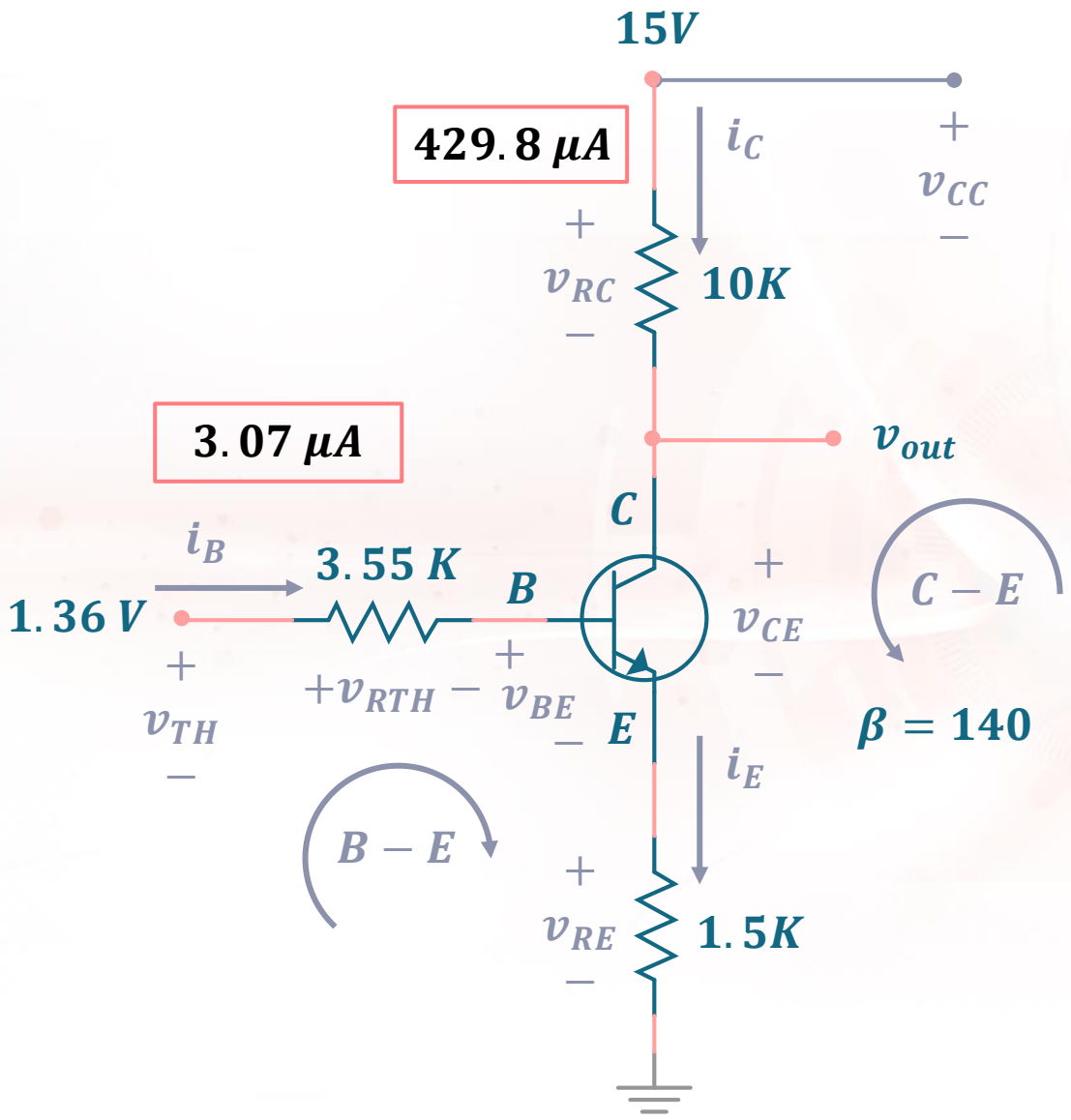
$$\beta = \frac{i_C}{i_B}$$

$$i_C = \beta i_B$$

$$i_C = 140(3.07\mu A)$$

$$\boxed{i_C = 429.8\ \mu A}$$

ans



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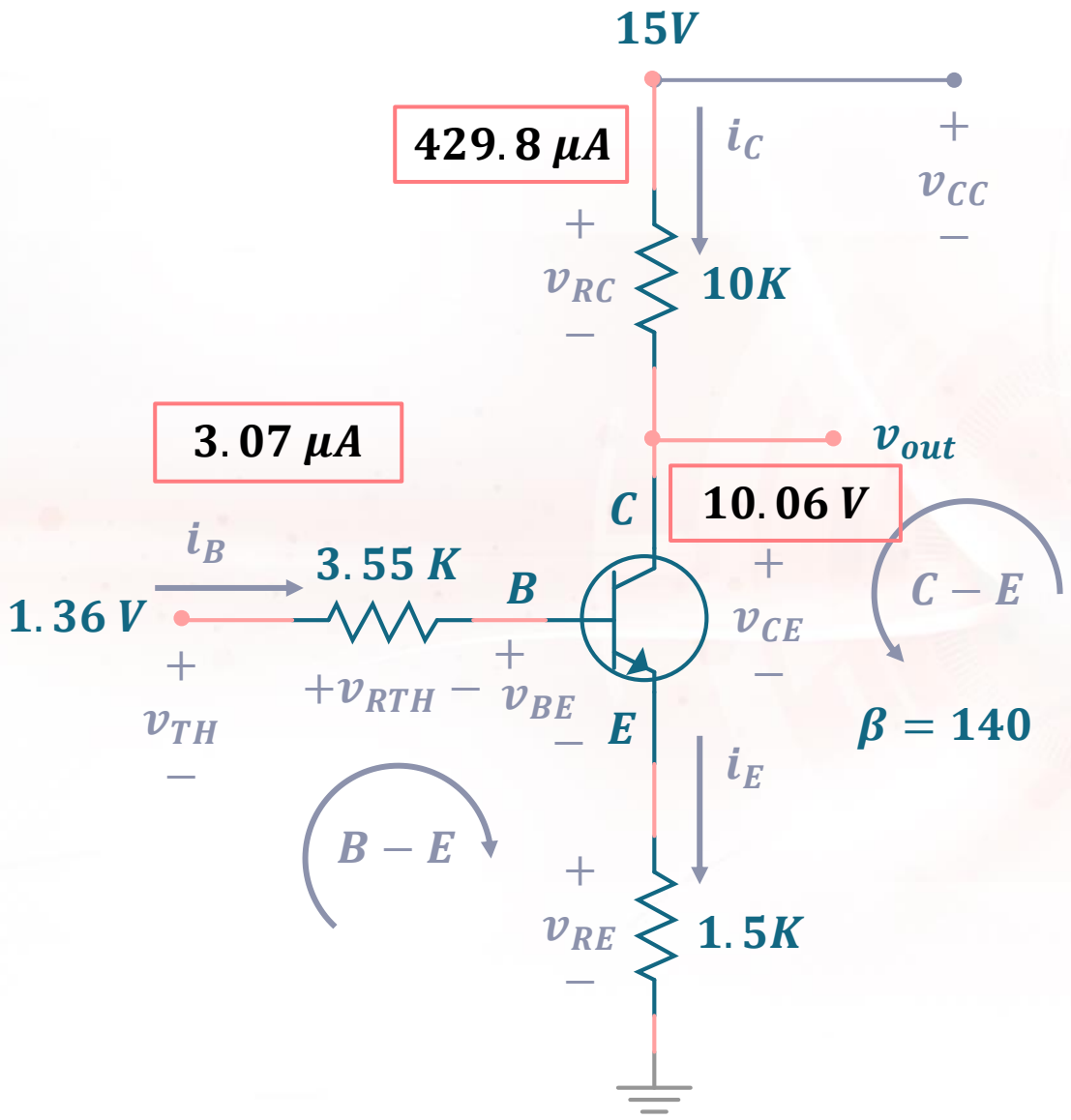
$$v_{CE} = v_{CC} - i_C R_C - i_E R_E$$

$$v_{CE} = v_{CC} - i_C(R_C + R_E)$$

$$v_{CE} = 15 - 429.8\mu A(10K + 1.5K)$$

$$\boxed{v_{CE} = 10.06\ V}$$

ans



KVL @$C - E$

$$-v_{CC} + v_{RC} + v_{CE} + v_{RE} = 0$$

$$v_{RE} = v_{CC} - v_{RC} - v_{CE}$$

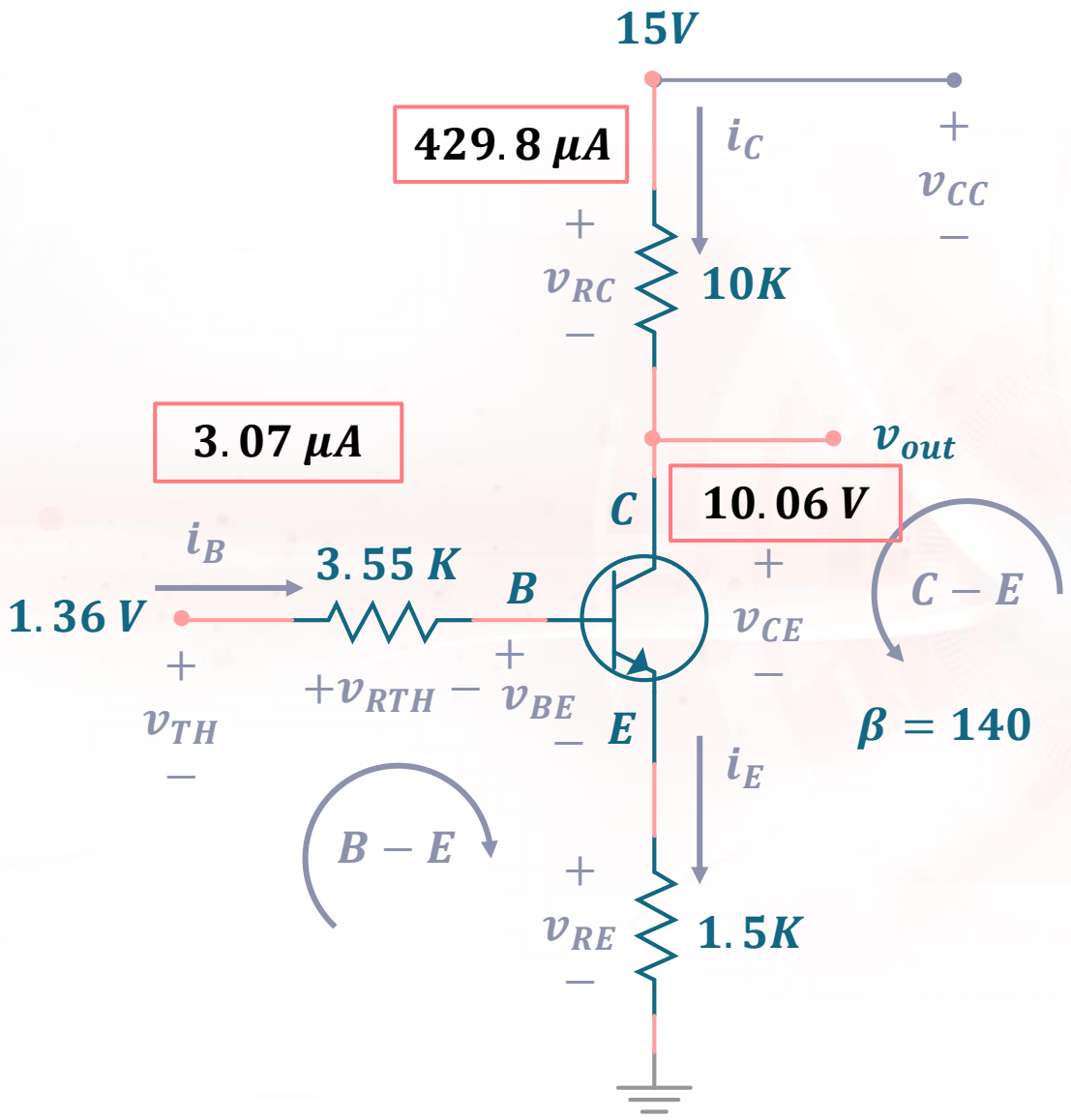
$$v_{RE} = v_{CC} - i_C R_C - v_{CE}$$

$$v_{RE} = 15 - 429.8\mu A(10K) - 10.06$$

$$\boxed{v_E = 642\ mV}$$

ans

v_{E-}



Let $i_C \approx i_E$

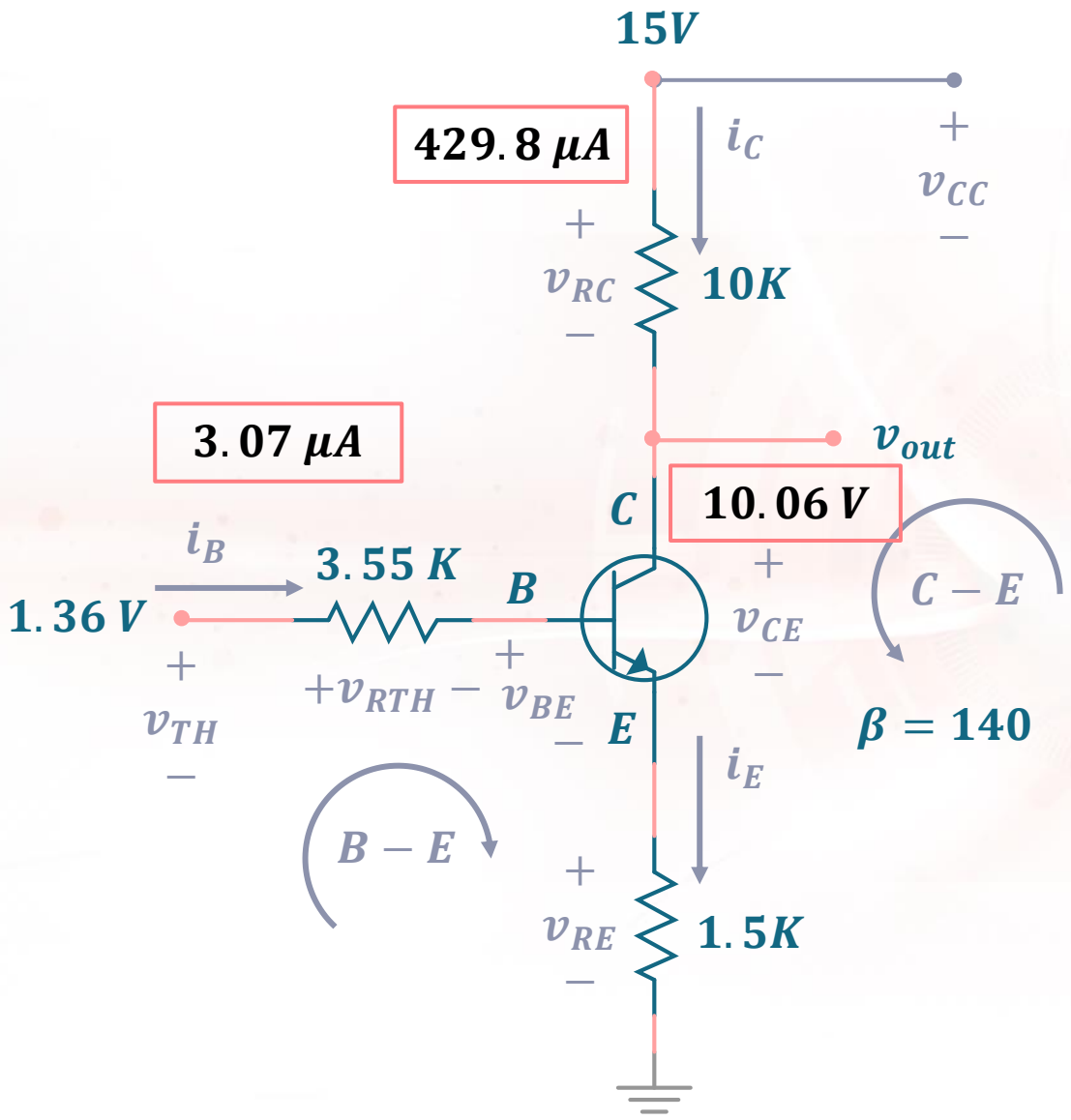
$$v_{RE} = i_E R_E$$

$$v_{RE} = 429.8\ \mu A(1.5K)$$

$$\boxed{v_E = 644.7\ mV}$$

ans

v_{E-}



$$i_E = i_B + i_C$$

$$i_E = 3.07\ \mu A + 429.8\ \mu A$$

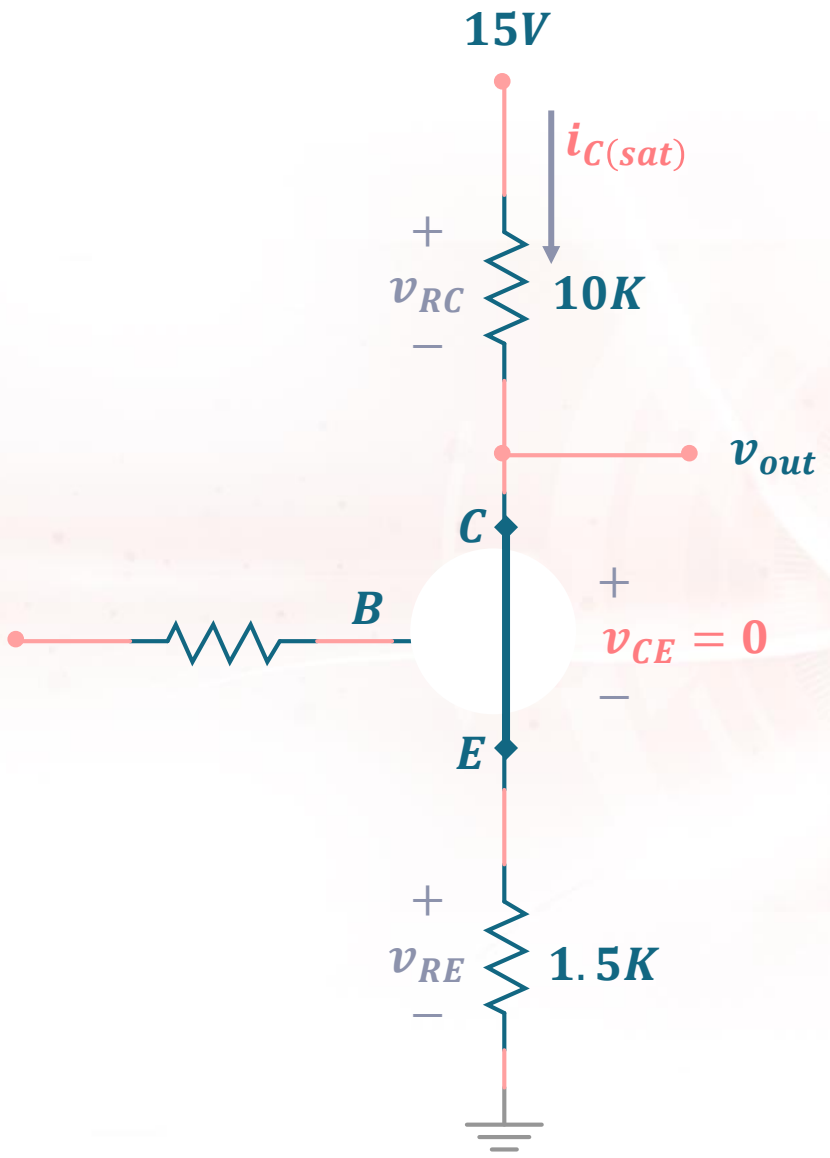
$$i_E = 432.87\ \mu A$$

$$v_{RE} = i_E R_E$$

$$v_{RE} = 432.87\ \mu A(1.5K)$$

$$\boxed{v_{RE} = 649.31\ mV}$$

ans



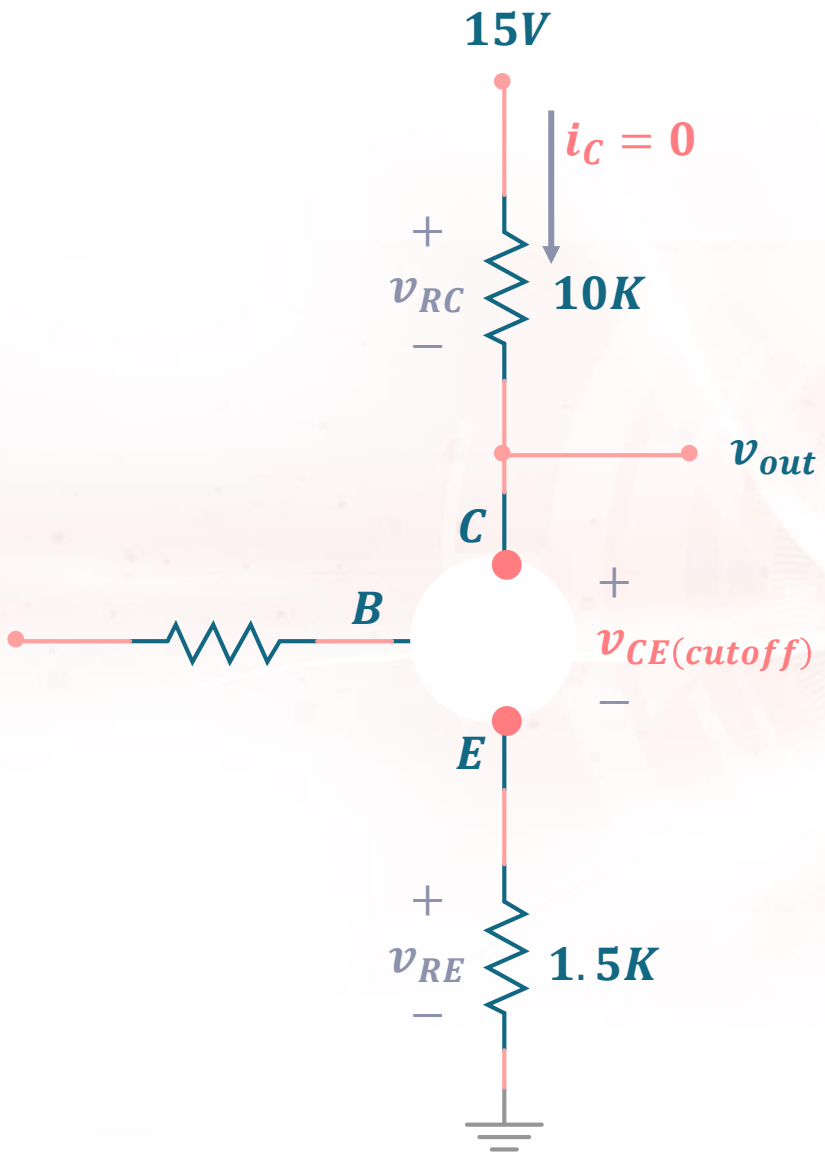
Mental Short,

$$i_{C(sat)} = \frac{v_{CC}}{R_C + R_E}$$

$$i_{C(sat)} = \frac{15}{10K + 1.5K}$$

$$\boxed{i_{C(sat)} = 1.3\ mA}$$

ans



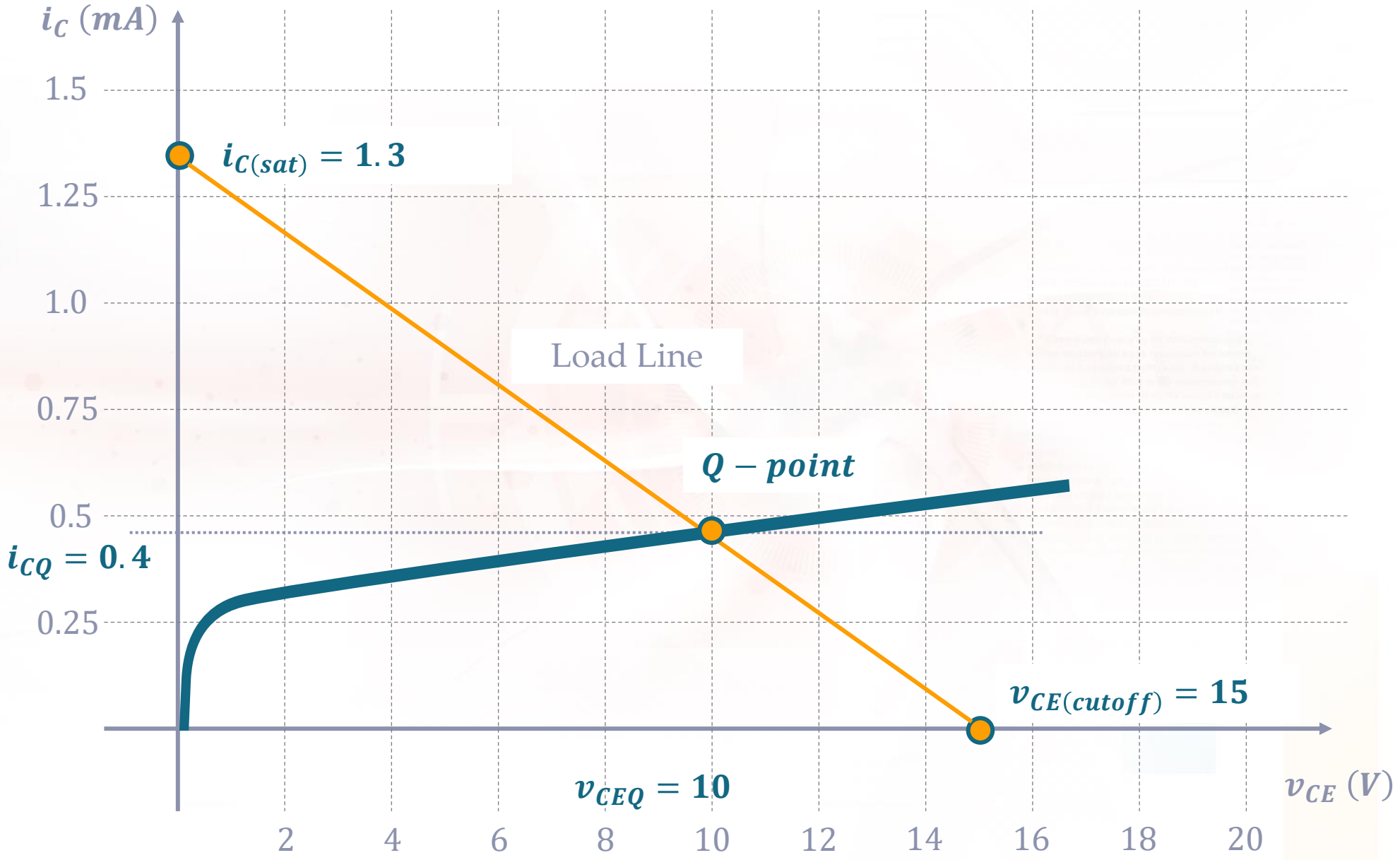
Mental Open,

$$v_{CE(cutoff)} = v_{CC}$$

$$\boxed{v_{CE(cutoff)} = 15\ V}$$

ans

LOAD LINE ANALYSIS



INDEPENDENT OF THE TRANSISTOR BETA

β	$i_C(mA)$	$V_{CE}(V)$	$\%\Delta V_{CE}$
140	0.85	12.2	2%
70	0.83	12.5	

LABORATORY